

Problem Categories

The following are categories of high-yield problems you may encounter on the exam. No claim is made that this list is exhaustive, or complete; rather it is based on the references the PEBC provides and our best opinion of what an entry level pharmacist should know

1. Problem Type 1 - Using density or specific gravity of a substance to find its mass or volume
2. Calculating quantities based on given ratios, percentages, or concentrations of active ingredients relative to their diluent
3. Calculating the dispensed amount of medication
4. Calculating dosing options
5. Using Young's rule, Clark's rule, Fried's rule, and Cowling's rule for converting adult doses to pediatric doses
6. Ideal Body Weight
7. Calculating a patient's body mass index (BMI)
8. Calculating a patient's body surface area (BSA)
9. Units of activity and other measures of potency
10. Estimate Creatinine Clearance
11. Estimate Iron Required
12. Heparin-Dosing calculations
13. Isotonicity, *i* factor, E-values, and freezing point calculations
14. Determining the pH of solutions
15. Buffers and buffer solutions
16. Calculating milliequivalents (mEq)
17. Calculating osmolarity
18. Determining the infusion rate of an intravenously administered drug
19. Calculating the amount of ingested calories
20. Calculating the amount or concentration of a stock solution required to make a solution of different strength or volume (for the same active ingredient)
21. Calculating quantities of ingredients for compounding a specified strength of a formulation, given different strengths of stock formulations using the alligation method
22. Compounding Calculations
23. Calculations involving plant extracts
24. Calculation of active drug moiety
25. Radiopharmaceutical calculations
26. Bioavailability Calculations
27. Volume of distribution Calculations
28. Drug elimination Calculations
29. Calculating Drug Therapy Costs
30. Absolute Risk Reduction, Number Needed to Treat, Relative Risk, Relative Risk Reduction
31. Determining the minimum weighable quantity (MWQ) of a balance
32. Calculating percentage of error
33. Chemotherapy Dosage regimens

Conversion Factors You Must Know

According to the PEBC Web Site:

"The answers in calculation questions and the majority of values presented will be in SI (metric) units. However, you may be required to convert from the avoirdupois or apothecary systems before performing your calculation in instances where the conversion factor should be commonly known by an entry-level pharmacist (e.g. pounds to kilograms)"

-Source: PEBC Web Site. Access from

http://www.pebc.ca/index.php/ci_id/3074/la_id/1/print/true/art_id/3144.htm

We will not get into the details of what constitutes the avoirdupois and apothecary systems of measure; however, the following are **conversion factors you should know** to convert from these systems into the metric system, which is also known as SI units (a French abbreviation for *Système International d'unités*).

Some common conversion factors (those thought to be most applicable to exam are bolded):

1 teaspoon = 5 mL
1 tablespoon = 15 mL
1 fluid ounce = 30 mL
1 pint = 16 fluid ounces
1 quart = 2 pints
1 gallon = 4 quarts
1 kg = 2.2 lbs
1 meter = 3.3 feet
1 meter = 39.4 inches
1 foot = 12 inches
1 grain = 65 mg

Some formulas:

$$\text{Density} = \frac{\text{mass}}{\text{volume}}$$

$$\text{Molecular mass} = \frac{\text{mass}}{\text{mol}}$$

$$pH = -\log [H^+]$$

$$pOH = -\log [OH^-]$$

$$pH = pK_a + \log \frac{[A^-]}{[HA]}$$

$$pOH = pK_b + \log \frac{[BH^+]}{[B]}$$

Where:

K_a	is the acid dissociation constant
K_b	is the base dissociation constant
$[HA]$	is the molar concentration of the undissociated weak acid
$[A^-]$	is the molar concentration of the acid's conjugate base
$[B]$	is the molar concentration of the base, B
$[BH^+]$	denotes molar concentration of the conjugate acid of B

Some Values:

Normalized Adult Body Surface Area = 1.73 m²

Example 1:

How many fluid ounces are there in 1.7L?

1 fluid ounce = 30 mL

$$\frac{1700 \text{ mL}}{30 \frac{\text{mL}}{\text{fl oz}}} = 56.7 \text{ fluid ounces}$$

Example 2:

An ancient doctor prescribed 5 grains of diphenhydramine for a patient. The molecular weight of diphenhydramine is 255 g/mol. Because you prefer to live in the present, you'd like to know how many mg of diphenhydramine he gave to the patient.

Cross-multiplication:

$$\frac{1 \text{ grain}}{65 \text{ mg}} = \frac{5 \text{ grains}}{x} \quad x = 325 \text{ mg}$$

Therefore, you need approximately 325mg to make 5 grains of diphenhydramine.

Problem types you may face of the exam

Problem Type 1 - Using density or specific gravity of a substance to find its mass or volume

Specific gravity is a ratio that expresses how many times heavier (or lighter) a liquid is than water. A liquid with a specific gravity of 2, for example, will be twice as heavy as water for the same volume.

Because specific gravity is a ratio of two masses; it is a number without units.

Knowing the specific gravity of a liquid will allow you to convert its mass into volume or vice-versa.

The following formula may be used for problems relating to specific gravity:

$$\text{Grams of Other Liquid} = \text{Grams of Equal Volume of Water} \times \text{Specific Gravity}$$

Example 3:

What is the weight of 0.5 L of an alcohol with specific gravity of 0.75?

Weight of equivalent volume of water = 500 g (since water's density is 1 mL/1 g)

$$\begin{aligned}\text{Grams of Alcohol} &= \text{Grams of Equal Volume of Water} \times \text{Specific Gravity} \\ &= 500 \text{ g} \times 0.75 \\ &= \mathbf{375 \text{ g}}\end{aligned}$$

Example 4:

15mL of *compound J* has a mass of 10g. Calculate *compound J's* specific gravity.

We know that 15mL of water has a mass of approximately 15g.

$$\text{specific gravity} = \frac{\text{mass of the compound}}{\text{mass of water occupying the same volume}} = \frac{10 \text{ g}}{15 \text{ g}} = \mathbf{0.67}$$

Example 5:

A new antineoplastic drug has a specific gravity of 1.42. The pharmaceutical company manufactures the formulation as 2mL ampoules. One of the cancer patients at the hospital receives 2 ampoules BID. Calculate the mass of the drug for a day's supply for the patient in question.

Number of ampoules the patient receives per day:

2 ampoules x 2 (because the directions were BID) = 4 ampoules

$$\text{Mass of drug per ampoule} = \frac{1.42 \text{ g}}{\text{mL}} \times \frac{2 \text{ mL}}{\text{ampoule}} = \mathbf{2.84 \text{ g/ampoule}}$$

*Note that although specific gravity has no units, it can be thought of as having the units of density when asked to calculate an entity that is directly dependent upon the specific gravity. Recall that specific gravity is just a ratio of two densities (that is the reason why the number is unitless).

Total mass of drug for a daily supply for our patient:

$$\frac{2.84 \text{ g}}{\text{ampoule}} \times 4 \text{ ampoules} = \mathbf{11.36 \text{ g}}$$

Example 6:

A pycnometer is a bottle used to measure specific gravity. When empty it weighs 25g. When filled with water it weighs 51g. When filled with liquid X it weighs 77g. What is the specific gravity of this liquid?

Weight of water in bottle	= Weight of bottle when full of water – weight of bottle when empty = 51 g – 25 g = 26 g
Weight of other liquid in bottle	= Weight of bottle full of other liquid – weight of bottle when empty = 77 g – 25 g = 52 g
Specific gravity	= Mass of other liquid / Mass of same volume of water = 52 g / 26 g = 2.0

This means for any given volume, this other liquid will weigh twice as much as water.

Problem Type 2 – Calculating quantities based on given ratios, percentages, or concentrations of active ingredients relative to their diluent

Percent w/w, w/v, v/v

% volume per volume (v/v) is expressed as mL per 100 mL of solvent

% weight per volume (w/v) is expressed as grams per 100 mL of solvent

% weight per weight (w/w) is expressed as grams of solute per 100 grams of diluent

Strengths of medications are often given in such formats, and understanding how to convert between these formats and the mass or volume is a useful skill set.

Strengths of very dilute solutions can be expressed in parts per million (ppm) or parts per billion (ppb). This describes the number of parts of an agent per 1 million or 1 billion of the whole.

Example 7:

A suspension of drug C is 23% w/v. What is the concentration of this suspension in g/L?

23% w/v means 23 g /100 mL.

Cross-multiplication:

$$\frac{23 \text{ g}}{100 \text{ mL}} = \frac{x}{1000 \text{ mL}}$$

$$x = 230\text{g}$$

Therefore, the concentration is **230 g/L**.

Example 8:

How many grams of NaCl is needed to make 2L of a 0.9% w/v solution?

0.9% w/v means 0.9 g of NaCl / 100mL

$$\frac{0.9 \text{ g}}{100 \text{ mL}} \times 2000 \text{ mL} = \mathbf{18 \text{ g}}$$

Example 9:

DL is a diabetic patient who took her blood glucose reading after a dose of insulin made her feel weak and dizzy. The reading indicates she is hypoglycemic, so DL decides to dissolve 2g of glucose in 50mL of water. State the % w/w of this solution.

50mL of water = 50g of water

Total mass = 50g water + 2g glucose = 52g

Cross-multiplication:

$$\frac{52 \text{ g}}{100\%} = \frac{2 \text{ g}}{x}$$
$$x = \frac{2 \text{ g} \times 100\%}{52 \text{ g}} = \mathbf{3.85\%}$$

Example 10:

The pharmacy supplier catalog lists a calcium suspension with a strength of 1:150. Calculate how much calcium would be in 10mL of this solution.

1:150 means 1g of calcium in 150mL of solution

Cross-multiplication:

$$\frac{1 \text{ g calcium}}{150 \text{ mL}} = \frac{x}{10 \text{ mL}}$$

$$x = \mathbf{0.067\text{g} = 67 \text{ mg of calcium}}$$

Example 11:

KL comes into the clinic with full nasal congestion. The doctor would like you to prepare 90mL of oxymetazoline 0.2mM solution for KL. Molecular weight of oxymetazoline = 260 g/mol.

In solving this problem you should note that M means “molar” – which means moles/litre.

a) How much oxymetazoline powder do you need to dissolve in your diluent (0.9% NaCl solution)?

$$0.2 \text{ mM} = 0.2 \text{ mmol/L} = 0.2 \text{ } \mu\text{mol/mL}$$

$$0.2 \frac{\mu\text{mol}}{\text{mL}} \times 90 \text{ mL} = 18 \text{ } \mu\text{mol}$$

$$\text{Molecular weight} = 260 \text{ g/mol} = 260 \text{ mg/mmol} = 0.26 \text{ mg/} \mu\text{mol}$$

$$\text{Mass}_{\text{oxymetazoline}} = 18 \text{ } \mu\text{mol} \times 0.26 \frac{\text{mg}}{\mu\text{mol}} = \mathbf{4.68 \text{ mg}}$$

b) You ran out of NaCl stock solution. How do you make the required diluent?

Cross-multiplication:

$$\frac{0.9 \text{ g}}{100 \text{ mL}} = \frac{x}{90 \text{ mL}} \quad \mathbf{x = 0.81 \text{ g or } 810 \text{ mg of NaCl}}$$

Therefore you need to add 810 mg of NaCl to (90-0.81mL) = 89.2 mL of water to produce the required diluent.

Example 12:

When reconstituting an antibiotic, you add water to 300mg of antibiotic powder in order to make a total of 150mL of suspension. What is the % w/v of antibiotic in the suspension?

$$300\text{mg} = 0.3\text{g}$$

Solve the ratio using cross-multiplication:

$$\frac{0.3 \text{ g}}{150 \text{ mL}} = \frac{x}{100 \text{ mL}} \quad \mathbf{x = 0.2 \text{ g}}$$

Therefore, the % w/v is 0.2

Example 13:

Express 0.04% v/v as a ratio (in the form of 1:x).

$$0.04 \% = \frac{0.04}{100}$$

But we'd like the ratio expressed in whole numbers

$$\frac{0.04}{100} = \frac{1}{x}$$

Solving for x gives:

$$x = \frac{1 \times 100}{0.04} = \mathbf{2500}$$

Therefore, the **ratio is 1:2500**.

Example 14:

The concentration of a drug additive in an animal feed is 23.8 ppm. How many milligrams of drug should be used to create 40.5 kg?

23.8 ppm = 23.8 g (drug) in 1,000,000 g (feed)

Solve using cross multiplication:

$$\frac{1,000,000 \text{ g}}{23.8 \text{ g}} = \frac{40,500 \text{ g}}{x \text{ g}} \quad \mathbf{x = 0.9639 \text{ g} = 963.9 \text{ mg}}$$

Problem Type 3 – Calculating the dispensed amount of medication

Calculating the number of puffs for a specific inhaler prescription allows the pharmacist to dispense the correct number of inhalers per specified day's supply. Similarly, the number of drops, volume of liquid, or other doses may need to be calculated when interpreting a prescription.

Example 15:

Mrs CD comes in to the pharmacy with a prescription for Symbicort® (formoterol 6mcg and budesonide 100 mcg) 1 puff BID. If there are 120 doses in a turbuhaler, how many turbuhalers do you need to dispense for a 3-month supply for Mrs CD?

$$\frac{2 \text{ puffs}}{\text{day}} \times 90 \text{ days} = \mathbf{180 \text{ puffs}}$$

$$\frac{180 \text{ puffs}}{120 \frac{\text{puffs}}{\text{turbuhaler}}} = \mathbf{1.5 \text{ turbuhalers}}$$

Since you cannot dispense half an inhaler, you round up.
Therefore, you must dispense 2 turbuhalers for a 90-day supply.

Example 16:

OP comes in to the pharmacy looking for cromolyn sodium eye drops to relieve her red eyes. The doctor told her to instill 1 gtt QID ou. OP is planning to go on vacation for 30 days. If the bottle of cromolyn sodium eye drops has 5mL, how many bottles does OP need to last her for 30 days? Assume there are 15 drops of cromolyn sodium per mL of solution.

$$1 \text{ bottle} = 5 \text{ mL} \times 15 \frac{\text{drops}}{\text{mL}} = 75 \text{ drops}$$

$$\text{Number of daily drops} = 1 \text{ drop} \times 2 \text{ eyes} \times 4 \frac{\text{times}}{\text{day}} = 8 \frac{\text{drops}}{\text{day}}$$

Cross multiply:

$$\frac{1 \text{ day}}{8 \text{ drops}} = \frac{30 \text{ days}}{x} \quad x = 240 \text{ drops}$$

$$\frac{240 \text{ drops}}{75 \frac{\text{drops}}{\text{bottle}}} = \mathbf{3.2 \text{ bottles}}$$

Therefore, she will need 4 bottles to last her for 30 days.

Example 17:

The oral dose of a drug is 3 mg. The solution contains 0.5% w/v of the drug in a dropper bottle that delivers 15 drops/mL, with a total volume of 30 mL. The drug is taken twice daily for 30 days. How many bottles need to be dispensed

0.5% w/v means 0.5 g drug/100 mL or 500 mg/100 mL

First, we find the number of drops per mL:

$$\frac{15 \text{ drops}}{\text{mL}} \div \frac{100 \text{ mL}}{500 \text{ mg}} = \frac{3 \text{ drops}}{\text{mg}}$$

Second, we find the amount of drops needed for 3 mg of the drug

$$\frac{3 \text{ drops}}{\text{mg}} \times 3 \text{ mg} = 9 \text{ drops per dose}$$

Third, since the patient needs 60 doses which is 18 drops per day we calculate the number of drops in the bottle and how many bottles are required.

$$1 \text{ bottle} = \frac{15 \text{ drops}}{\text{mL}} \times 30 \text{ mL} = 450 \text{ drops}$$

Cross multiply

$$\frac{1 \text{ day}}{18 \text{ drops}} = \frac{30 \text{ days}}{x} \quad x = 540 \text{ drops}$$

$$\frac{540 \text{ drops}}{450 \frac{\text{drops}}{\text{bottle}}} = \mathbf{1.2 \text{ bottles}}$$

Therefore, the patient will need 2 bottles to last them 30 days.

Problem type 4 – Calculating dosing options

Pharmacists will encounter dosing options that differ from a usual dosage. Low-dose and high-dose therapies, fixed-dose combination products, tablet splitting, and special dosing regimens are common and may need to be calculated in practice.

Example 18:

If a patient is changed from a standard dose product containing 0.85 mg estrogen to a low-dose formulation containing 0.35 mg estrogen, how many milligrams less of estrogen would the patient take per week.

$$\begin{aligned}0.85 - 0.35 \text{ mg} &= 0.50 \text{ mg} \\0.5 \text{ mg} \times 7 \text{ days} &= \mathbf{3.5 \text{ mg}}\end{aligned}$$

Therefore, the patient will be taking 3.5 mg of estrogen less per week.

Example 19:

A patient attempted to split a 30 mg tablet in half, resulting in 'half tablets' differing by 2.2 mg in drug content. Assuming a whole tablet has uniform drug content, calculate the amount of drug in each 'half' tablet.

If L = large half and S = smaller half

Then $L + S = 30$ and $L - S = 2.2$

$$\begin{aligned}L - (30 - L) &= 2.2 \\2L &= 32.2 \text{ mg}\end{aligned}$$

$$\mathbf{L = 16.1 \text{ mg}}$$

$$\mathbf{S = 30 - 16.1 = 13.9 \text{ mg}}$$

Therefore, the larger 'half' has 16.1 mg and the smaller half has 13.9 mg of drug.

Example 20:

A 12-week regimen for a brand of nicotine patch calls for a patient to wear a 15-mg patch each day for the first 4 weeks, a 10-mg patch each day for the next 2 weeks, and a 5-mg patch each day for the remainder of the time. Overall, how many milligrams of nicotine are administered?

$$15 \text{ mg} \times 28 + 10 \text{ mg} \times 14 + 5 \text{ mg} \times 42 = \mathbf{770 \text{ mg}}$$

Therefore, 770 mg of nicotine is administered over the 12-week regimen.

Problem Type 5 – Using Young’s rule, Clark’s rule, Fried’s rule, and Cowling’s rule for converting adult doses to pediatric doses

You may be asked to convert adult doses to pediatric doses in four ways. The first method, Young’s rule, is based on age, second method, Clark’s rule, is based on mass, the third method, Cowling’s rule, is based on the age at next birthday, and the fourth method, Fried’s rule is based on age in months. In practice, these dosing rules are only used if a pediatric dosing reference is not available (which is rare). You may be given only the information required to use one of these equations (either age or weight). In cases where information is provided that allows you to use all four rules, it may be a good idea to select the more conservative answer, which would be the smaller of the four.

Example 21:

HX is an 8-month-old infant who weighs 10kg. He visits the doctor, as his mom noticed the spring pollen is causing significant rhinorrhea and sneezing in her child. The doctor remembers the usual dose of diphenhydramine for an average adult is 25mg q4-6h. He asks you to calculate the pediatric dose for HX using Young’s rule, Clark’s rule, Fried’s rule, and Cowling’s rule, and choose the smaller of the four. Calculate the dose of diphenhydramine for HX.

$$8 \text{ months} \times 1 \text{ year}/12 \text{ months} = 0.67 \text{ years}$$

$$10 \text{ kg} \times 2.2\text{lbs/kg} = 22 \text{ lbs}$$

Young’s Rule: (age based)

$$\text{Pediatric dose} = \text{dose for average adult} \times \frac{\text{age in years}}{(\text{age in years} + 12)} = 25 \text{ mg} \times \frac{0.67}{12.67} = \mathbf{1.32 \text{ mg}}$$

Clark’s Rule: (mass based)

$$\text{Pediatric dose} = \text{dose for average adult} \times \frac{\text{mass of child in lbs}}{150} = 25 \text{ mg} \times \frac{22}{150} = \mathbf{3.67 \text{ mg}}$$

Fried’s Rule: (age in months based)

$$\text{Pediatric dose} = \text{dose for average adult} \times \frac{\text{age in months}}{150} = 25 \text{ mg} \times \frac{8}{150} = \mathbf{1.33 \text{ mg}}$$

Cowling’s Rule: (age in years at next birthday)

$$\text{Pediatric dose} = \text{dose for average adult} \times \frac{\text{age at next birthday}}{24} = 25 \text{ mg} \times \frac{1}{24} = \mathbf{1.04 \text{ mg}}$$

Therefore, the diphenhydramine dose the doctor wishes to provide for HX is 1.04 mg q4-6h.

Problem Type 6 – Ideal Body Weight

Ideal body weight formulas are dependent upon the gender and the height of the patient.

For women:

$$45.5 \text{ kg} + 2.3 \text{ kg} \times \text{number of inches above 5 feet in height}$$

For men:

$$50 \text{ kg} + 2.3 \text{ kg} \times \text{number of inches above 5 feet in height}$$

Reference:

Calculator.net. *Ideal Weight Calculator*. Accessed April 2014. <http://www.calculator.net/ideal-weight-calculator.html>.

Problem Type 7 – Calculating a patient's body mass index (BMI)

This is one of the most common calculations used in clinical pharmacy. A person's BMI is based on the patient's height and mass. Apart from bodybuilders and pregnant women, BMI accurately categorizes patients according to their optimal weight range. Patients outside of their optimal weight range are at increased risk of certain medical conditions.

Example 22:

FV has been staying on track with his New Year resolution to lose weight. He exercises regularly and consumes less high-calorie soft drinks. He comes to the pharmacy to ask you to help him calculate his BMI. FV is 5 foot 5 inches tall and currently weighs 160 lbs.

$$160 \text{ lbs} \div 2.2 \text{ lbs/kg} = 72.7 \text{ kg}$$

$$5 \text{ foot } 5 \text{ inches} = 65 \text{ inches}$$

$$65 \text{ inches} \div 39.4 \text{ inches/m} = 1.65 \text{ m}$$

$$BMI = \frac{\text{mass in kg}}{(\text{height in m})^2} = \frac{72.7 \text{ kg}}{(1.65 \text{ m})^2} = 26.7 \text{ kg/m}^2$$

Therefore, FV is considered overweight (BMI between 25 – 30 kg/m²).

Problem Type 8 – Calculating a patient's body surface area (BSA)

Most BSA calculations are done using a nomogram, but it may also be determined using the Mosteller Formula:

$$BSA, m^2 = \sqrt{\frac{\text{Height (cm)} \times \text{Weight (Kg)}}{3600}}$$

There is also the normalized adult BSA: 1.73 m².

Example 23:

Calculate the BSA for a patient weight 73 kg and measuring 180 cm.

$$BSA, m^2 = \sqrt{\frac{180 \text{ cm} \times 73 \text{ kg}}{3600}} = 1.91 m^2$$

Therefore, the patients BSA is 1.91 m².

Problem Type 9 – Units of activity and other measures of potency

Certain medications potency such as antibiotics, endocrine products, vitamins, and biologics are based on activity and are expressed as units of activity, microgram per milligram, or in other standardized terms of measurement. The activity of a drug has no relation to the activity of another drug.

A common drug for which potency is expressed in units is insulin. All insulin is standardized to contain either 100 (U -100) or 500 (U- 500) units per milliliter.

The strength of viral vaccines are commonly expressed in terms of tissue culture infectious dose (TCID₅₀), which is the estimated amount of virus required to infect 50% of inoculated cultures. Toxoids are expressed in terms of flocculating units (Lf Unit), where 1 Lf unit has the capacity to flocculate one unit of standard antitoxin.

Example 24:

How many milliliters of U-500 should be used to obtain 60 units of insulin?

U-500 insulin contains 500 units/mL

Cross multiply:

$$\frac{500 \text{ units}}{60 \text{ units}} = \frac{1 \text{ mL}}{x \text{ mL}} \quad x = 0.12 \text{ mL}$$

Therefore, 0.12 mL of U-500 insulin is required to obtain 60 units of insulin.

Example 25:

A biologic contains 80 Lf Units of toxoid Y in each 3.5 mL of product. If a patient is set to receive 15 Lf Units, how many milliliters of the product should be given?

Cross multiply:

$$\frac{80 \text{ Lf Units}}{15 \text{ Lf Units}} = \frac{3.5 \text{ mL}}{x \text{ mL}} \quad x = 0.66 \text{ mL}$$

Therefore, 0.66 mL of the product is required to give the patient 15 Lf units.

Problem Type 10 – Estimate Creatinine Clearance

The Cockcroft-Gault formula is an estimation of a patient's renal function. It was derived by using male patients, thus there is no real data for female patients. As such, the entire formula is multiplied by a factor of 0.85 if used to estimate renal function in a female. The lean body mass of the patient must also be estimated when using the formula. One of the limitations of the Cockcroft-Gault formula is that it can either overestimate the glomerular filtration rate (GFR) in patients who have rapidly-declining renal function or it may underestimate the GFR in patients who are rapidly recovering from an illness/condition that initially lowered their renal function. As seen in the formula, the Cockcroft-Gault equation estimates renal function by taking into consideration the patient's age, gender, ideal body mass, and serum creatinine level.

$$CrCl \text{ (mL/min)} = \frac{1.2 \times (140 - \text{age}) \times \text{ideal body weight (kg)}}{\text{serum creatinine} \left(\frac{\mu\text{mol}}{\text{L}} \right)} \text{ (multiply by 0.85 if female)}$$

$$CrCl \text{ (mL/min)} = \frac{(140 - \text{age}) \times \text{ideal body weight (kg)}}{72 \times \text{serum creatinine} \left(\frac{\text{mg}}{\text{dL}} \right)} \text{ (multiply by 0.85 if female)}$$

It is also sometimes desirable to adjust the calculated creatinine clearance for body surface area to account for possible variations in drug dosages. The formula to adjust creatinine clearance for body surface area is as follows:

$$\text{Adjusted CrCl} \left(\frac{\text{mL}}{\text{min}} \right) = \frac{\text{BSA (m}^2\text{)}}{1.73\text{m}^2} \times CrCl$$

Example 26:

A 65-year-old gentleman is having his annual physical exam. He weighs 70 kg, is 5 foot 10 inches (1.78 m) and his blood work shows that he has a serum creatinine of 1.2 mg/dL. a) Using the Cockcroft-Gault formula estimate his creatinine clearance rate. b) What is his adjusted creatine clearance based on body surface area.

a) First find ideal body weight:

$$50 \text{ kg} + 2.3 \text{ kg} \times 10 = \mathbf{73 \text{ kg}}$$

As actual body weight < ideal body weight, we will use actual body weight.

Next, solve for CrCl:

$$CrCl \left(\frac{\text{mL}}{\text{min}} \right) = \frac{(140 - 65) \times 70 \text{ kg}}{72 \times 1.2 \frac{\text{mg}}{\text{dL}}} = \mathbf{60.76 \frac{\text{mL}}{\text{min}}}$$

Therefore, his estimated creatinine clearance rate is 60.76 mL/min.

b) Find BSA:

$$BSA, m^2 = \sqrt{\frac{178 \text{ cm} \times 70 \text{ Kg}}{3600}} = 1.86 \text{ m}^2$$

Solve for adjusted CrCl:

$$\text{Adjusted CrCl} \left(\frac{\text{mL}}{\text{min}} \right) = \frac{1.86 \text{ m}^2}{1.73 \text{ m}^2} \times 60.76 \frac{\text{mL}}{\text{min}} = 65.33 \frac{\text{mL}}{\text{min}}$$

Therefore, his adjusted creatinine clearance is 65.33 mL/min.

Problem Type 11 – Estimate Iron Required

Certain situations require specific formulas to determine a patients requirements. For example, there may be instances where the amount of iron required to bring hemoglobin back to normal is needed to be calculated.

$$\text{Iron required (mg)} = \text{Body weight (lb)} \times 0.3 \times \left[100 - \frac{\text{Hb} \left(\frac{\text{g}}{\text{dL}} \right) \times 100}{14.8 \frac{\text{g}}{\text{dL}}} \right]$$

Where 14.8 g/dL is the normal adult hemoglobin value and 0.3 is iron content percent.

Problem Type 12 – Heparin-Dosing calculations

Heparin is a polysaccharide that has anticoagulant properties and slows clotting time. Heparin, although is useful, carries a risk of hemorrhage. Salt forms of heparin are standardized to contain 180 USP Heparin Units per milligram. It is very important to calculate heparin dosing accurately due to the distinct risk heparin use carries.

Example 27:

An intravenous infusion contains 30,000 units of heparin sodium in 1500 mL of D5W. The rate of infusion is set to 1200 units/h for a 160-lb patient. Calculate (a) the concentration of heparin sodium in the infusion, in units/mL; (b) the length the infusion would run; (c) the dose of heparin administered, in unit/kg/min.

$$(a) \frac{30,000 \text{ units}}{1500 \text{ mL}} = 20 \frac{\text{units}}{\text{mL}}$$

$$(b) \frac{30,000 \text{ units}}{1200 \text{ units/h}} = 25 \text{ hours}$$

$$(c) 160 \text{ lb} = 72.7 \text{ kg}$$

$$25 \text{ hours} = 1500 \text{ minutes}$$

$$\frac{30,000 \text{ units}}{1500 \text{ minutes}} \div 72.7 \text{ kg} = 0.28 \text{ units/kg/min}$$

Problem Type 13 – Isotonicity, *i* factor, E-values, and freezing point calculations

Isotonic solutions are two solutions that have the same osmotic pressure across a semipermeable membrane, allowing for the free movement of water across the membrane. Ophthalmic preparations are formulated to be isotonic to lacrimal fluid for comfort, reduce irritation, and maintain stability. Most conveniently, freezing points are used to determine isotonicity and the freezing point of bodily fluids (blood serum and lacrimal fluid) is -0.52°C .

Electrolytes in weak solutions do not fully dissociate, the dissociation factor *i* (which tells us how many more particles are yielded per molecule) is required to find tonicity. Sodium chloride is commonly used as a reference point, where a 0.9% w/v solution is considered isotonic with body fluids.

Simple isotonic solutions can be calculated with this formula:

$$\frac{0.52 \times \text{molecular weight}}{1.86 \times \text{dissociation factor } (i)} = \text{g of solute per 100 g of water}$$

The dissociation factor of medicinal salts can be approximated if the number of ions is known and no other information is given:

1. Nonelectrolytes and substances with slight dissociation: 1.0
2. Substances that dissociate into 2 ions: 1.8
3. Substances that dissociate into 3 ions: 2.6
4. Substances that dissociate into 4 ions: 3.4
5. Substances that dissociate into 5 ions: 4.2

A special problem arises if the prescription requires the creation of an isotonic solution of a medication using a tonicity agent (such as sodium chloride). The amount of sodium chloride required depends on how much sodium chloride is represented by the medication. This is known as sodium chloride equivalents (E-values).

$$\text{Sodium Chloride equivalent} = \frac{\text{Molecular weight of sodium chloride}}{i \text{ factor of sodium chloride}} \times \frac{i \text{ factor of the substance}}{\text{molecular weight of substance}}$$

Isotonic solutions can also be prepared by volume adjustment. By adding purified water to a 1-g quantity of a drug with a known E-value, a calculated volume of an isotonic solution can be prepared. This solution can be diluted with an isotonic vehicle, such that the strength can be reduced while the solution remains isotonic.

Freezing point data (DT_f) can be used in isotonic calculations. The freezing point must be lowered to -0.52°C , so that the pharmaceutical agent can be considered isotonic.

Example 28:

Zinc chloride is a 3-ion electrolyte, dissociating 80% in a certain concentration. Calculate its dissociation (i) factor.

On the basis of 80% dissociation, 100 particles of zinc chloride will yield:

80 zinc ions

160 chloride ions

20 undissociated particles

Total of 260 particles

$$\frac{260}{100} = \mathbf{2.6}$$

Therefore, the dissociation factor is 2.6.

Example 29:

Calculate the sodium chloride equivalent of tropicamide, a nonelectrolyte with a molecular weight of 284.

Due to being a nonelectrolyte, tropicamide i factor = 1.0

$$\frac{58.5}{1.8} \times \frac{1}{284} = \mathbf{0.11}$$

The sodium chloride equivalent of tropicamide is 0.11.

Example 30:

How many grams of sodium chloride should be used in compounding the following prescription?

Moxifloxacin hydrochloride 0.7 g
Sodium Chloride qs
Purified water 50 mL
Make isotonic sol.
Sig. for the eye

Moxifloxacin hydrochloride has an E-value of 0.13 (determined experimentally)

Step 1: Determine how much sodium chloride is represented by moxifloxacin hydrochloride

$$0.7 \text{ g} \times \frac{1000 \text{ mg}}{1 \text{ g}} \times 0.13 = \mathbf{91 \text{ mg}}$$

91 mg of sodium chloride is represented by moxifloxacin hydrochloride.

Step 2: Determine how much sodium chloride is required to make the solution isotonic with tears

$$50 \text{ mL} \times \frac{0.9 \text{ g}}{100 \text{ mL}} \times \frac{1000 \text{ mg}}{1 \text{ g}} = 450 \text{ mg}$$

450 mg of sodium chloride in 50 mL of an isotonic solution

Step 3: Find the difference between the values from step 2 and step 1:

$$450 \text{ mg} - 91 \text{ mg} = 359 \text{ mg}$$

Therefore, 359 mg of sodium chloride is to be used.

Example 31:

How many grams of potassium nitrate (E-value of 0.58) should be used in compounding the following prescription?

Ingredient X	0.4 g
Pilocarpine hydrochloride	2 %
Potassium nitrate	qs
Purified water	70 mL
Make isotonic solution with tears	

Assume ingredient X is a new substance with a molecular weight of 376 and an i factor of 2.7.

$$\frac{58.5}{1.8} \times \frac{2.7}{376} = 0.23$$

The E-value of ingredient X is 0.23.

Step 1: Determine how much sodium chloride is represented by ingredient X and pilocarpine hydrochloride.

Ingredient X:

$$0.4 \text{ g} \times \frac{1000 \text{ mg}}{1 \text{ g}} \times 0.23 = 92 \text{ mg}$$

Pilocarpine hydrochloride:

$$\frac{2 \text{ g}}{100 \text{ mL}} \times \frac{1000 \text{ mg}}{1 \text{ g}} \times 70 \text{ mL} = 1400 \text{ mg} \times 0.24 = 336 \text{ mg}$$

Step 2: Determine how much sodium chloride is required to make the solution isotonic with tears

$$70 \text{ mL} \times \frac{0.9 \text{ g}}{100 \text{ mL}} \times \frac{1000 \text{ mg}}{1 \text{ g}} = 630 \text{ mg}$$

Step 3: Find the difference between the values from step 2 and step 1

$$630 \text{ mg} - 336 \text{ mg} - 92 \text{ mg} = \mathbf{202 \text{ mg}}$$

202 mg of sodium chloride is needed to make the solution. However the question is asking for potassium nitrate not sodium chloride.

$$202 \text{ mg} \div 0.58 = \mathbf{348.28 \text{ mg}}$$

Therefore, 348.28 mg of potassium nitrate is required to make the solution isotonic.

Example 32:

If silver nitrate has an E-value 0.33, (a) how many milliliters of isotonic solution may be prepared from 1 g of silver nitrate, and (b) how many milliliters of 0.9% w/v sodium chloride solution may be added to the resultant solution to prepare an isotonic solution having a 2% w/v concentration of silver nitrate?

(a) Cross multiply;

$$\frac{0.33 \text{ g}}{x \text{ mL}} = \frac{0.9 \text{ g}}{100 \text{ mL}} \quad \mathbf{x = 36.67 \text{ mL}}$$

Therefore, 36.67 mL of isotonic solution can be prepared with 1 g of silver nitrate.

(b) Use $Q1C1 = Q2C2$ to solve

First, calculate concentration of silver nitrate in 36.67 mL

$$\frac{1 \text{ g}}{36.67 \text{ mL}} \times 100\% = \mathbf{2.7\% \text{ w/v}}$$

Then apply the above equation

$$Q2 = \frac{36.67 \times 2.7\%}{2\%} = \mathbf{50 \text{ mL}}$$

Then solve for volume of 0.9% w/v sodium chloride is needed:

$$50 \text{ mL} - 36.67 \text{ mL} = \mathbf{13.33 \text{ mL}}$$

Therefore, 50 mL of solution can be prepared and 13.33 mL of 0.9% w/v sodium chloride is needed.

Example 33:

How many milligrams each of sodium chloride and morphine sulfate are required to prepare 45 mL of a 1% solution of morphine sulfate isotonic with tears? The DT_f of sodium chloride is 0.58 and the DT_f of morphine sulfate is 0.08.

To make the solution isotonic, the freezing point must be lowered to -0.52°C . The solution must be lowered to that temperature. We know that 1% morphine sulfate has a freezing point lowering of 0.08°C

$$0.52^\circ\text{C} - 0.08^\circ\text{C} = 0.44^\circ\text{C}$$

Then find the percent of sodium chloride needed to lower the temperature by an additional 0.44°C .

$$\frac{1\% \text{ NaCl}}{x\% \text{ NaCl}} = \frac{0.58^\circ\text{C}}{0.44^\circ\text{C}} \quad x = 0.76\%$$

Thus to make 45 mL of solution

$$\begin{aligned} \text{Morphine sulfate} &= 45 \text{ mL} \times 1\% = 0.45 \text{ g} = 450 \text{ mg} \\ \text{Sodium chloride} &= 45 \text{ mL} \times 0.76\% = 0.342 \text{ g} = 342 \text{ mg} \end{aligned}$$

Therefore, 450 mg of morphine sulfate and 342 mg of sodium chloride is required to prepare 45 mL of a 1% solution isotonic with tears.

Problem Type 14 – Determining the pH of solutions

You may be asked to calculate the pH of solutions to be administered to a patient. These are important when compared to the pH of bodily fluids or organs because values of pH that are either too high or too low can lead to significant adverse effects, as proteins in the body begin to denature when subjected to different hydronium (H_3O^+) concentrations. Moreover, the blood's buffering capacity may be compromised in cases where excessive OH^- or H_3O^+ ions are produced or introduced.

Example 34:

A solution of a certain amino acid has a pK_a of 3.8. The ratio of ionized to un-ionized amino acid in the solution is 120:1. What is the pH of this solution?

HA refers to the acid (un-ionized)

A^- refers to the base (ionized)

Ratio of HA to $\text{A}^- = 1/120 = 0.0083$

To solve use the Henderson–Hasselbalch equation:

$$pH = pK_a - \log \frac{[HA]}{[A^-]} = 3.8 - \log 0.0083 = 3.8 - (-2.1) = 5.9$$

Note: The above equation is not provided in the PEBC formula sheet, so you should memorize it.

Example 35:

A solution of magnesium carbonate has a carbonate ion concentration of 5M and a bicarbonate concentration of 2M. The K_b of carbonate is 2.1×10^{-4} . Assuming the magnesium ions do not contribute to the acidity or basicity, what is this solution's pH?

$$pK_b = -\log K_b = -\log 2.1 \times 10^{-4} = 3.68$$

$$\frac{[HB^+]}{[B]} = \frac{2\text{ M}}{5\text{ M}} = 0.4$$

$$pOH = pK_b + \log \frac{[HB^+]}{[B]} = 3.68 + \log 0.4 = 3.68 - 0.40 = 3.28$$

$$pH = 14 - pOH = 14 - 3.28 = 10.72$$

Note: The above equations are not provided in the PEBC formula sheet, thus you will have to memorize them.

Problem type 15 – Buffers and buffer solutions

A buffer solution is an aqueous solution that consists of a mixture of a weak acid and its conjugate base, or a weak base and its conjugate acid. The pH of buffer solutions does not change much when a small amount of strong acid or base is added. Buffer solutions are used in pharmacy to prepare certain dosage forms, manufacture of formulations that must maintain a stable pH, and for certain assays that require an adjustment/maintenance of a specific pH.

The buffer equation for a weak acid is:

$$pH = pK_a + \log \frac{\text{salt}}{\text{acid}}$$

The buffer equation for a weak base is:

$$pH = pK_w - pK_b + \log \frac{\text{base}}{\text{salt}}$$

Example 36:

What is the pH of a buffer solution prepared with 0.1 M sodium borate and 0.025 M boric acid? The pK_a value of boric acid is 9.24 at 25 °C.

$$pH = 9.24 + \log \frac{0.1}{0.025} = 9.84$$

The pH of the buffer is 9.84.

Example 37:

Calculate the change in pH after adding 0.07 mol of sodium acetate to a liter of a buffer solution containing 0.3 M concentrations each of sodium acetate and acetic acid. The pKa value of acetic acid is 4.76 at 25 °C.

First, find the pH of the buffer solution:

$$pH = pKa + \log \left(\frac{[A^-]}{[HA]} \right)$$

$$pH = 4.76 + \log \frac{0.3}{0.3} = 4.76$$

Sodium acetate is the conjugate base. Adding it increases the numerator in the Henderson-Hasselbalch equation. New moles of sodium acetate: 0.37 mol.

$$pH = 4.76 + \log \frac{0.37}{0.3} = 4.85$$

The pH of the buffer solution is 4.85.

$$4.85 - 4.76 = 0.09$$

The change in pH is 0.09.

Problem Type 16 – Calculating milliequivalents (mEq)

Knowing the number of milliequivalents (mEq) in ionic solutions allows clinicians to determine the total number of charges present. If the charges are imbalanced, the clinicians may then choose to add or remove certain cations or anions from the solution to achieve electrical neutrality. When compounding an electrolyte solution for a dehydrated patient, knowing the milliequivalent balance can help determine the appropriate proportions of electrolytes to be added to the solution.

$$mEq = \frac{\text{mass in mg of an ion} \times \text{valence of ion}}{\text{molecular mass in g/mol}}$$

You will need to memorize the above equation

Example 38:

An over-the-counter antacid formulation contains 5mg of $Al(OH)_3$ and 10mg of $CaCO_3$ (aluminum hydroxide and calcium carbonate). Calculate the number of milliequivalents (mEq) of Al^{3+} and Ca^{2+} .

The valence of Ca is 2 and the valence of Al is 3.

$$mEq = \frac{\text{mass in mg} \times \text{valence}}{\text{molecular mass in g/mol}} = \text{millimoles of an ion} \times \text{its valence}$$

For Al(OH)_3

Molecular mass = 78 g/mol

$$mEq = \frac{5 \text{ mg} \times 3}{78 \text{ g/mol}} = 0.19 \text{ mEq}$$

For CaCO_3

Molecular mass = 100 g/mol

$$mEq = \frac{10 \text{ mg} \times 2}{100 \text{ g/mol}} = 0.2 \text{ mEq}$$

Example 39:

KR has a history of anemia secondary to heavy menstruation. She takes 10mL BID of an iron supplement suspension containing 5 mEq/L of Fe^{3+} . What mass of Fe^{3+} does KR receive in a daily dose of her supplement?

KR receives a daily dose of:

$$5 \frac{\text{mEq}}{\text{L}} \times 0.02 \text{ L} = 0.1 \text{ mEq}$$

Molecular mass of Fe = 56 g/mol

$$\text{equivalent molecular mass of } \text{Fe}^{3+} = \frac{56 \frac{\text{g}}{\text{mol}}}{\text{valence}} = \frac{56 \frac{\text{g}}{\text{mol}}}{3} = 18.7$$

$$1 \text{ mEq } \text{Fe}^{3+} = \frac{18.7}{1000} = 0.0187 \text{ g or } 18.7 \text{ mg}$$

Solve for X via cross-multiplication:

$$\frac{1 \text{ mEq of } \text{Fe}^{3+}}{18.7 \text{ mg}} = \frac{0.1 \text{ mEq of } \text{Fe}^{3+}}{x} \quad x = 1.87 \text{ mg}$$

Therefore, KR receives a total daily dose of 1.87mg of elemental iron from her supplement.

Example 40:

What is the equivalent weight of the Mg^{2+} ion?

Molecular mass of Mg = 24.3 g/mol

$$\text{equivalent weight} = \frac{\text{molecular mass}}{\text{valence}} = \frac{24.3}{2} = 12.15$$

Example 41:

RT loves to consume huge amounts of unhealthy “food”, despite your valiant efforts to dissuade him. After winning a hot-dog-eating contest, he shows up to the pharmacy and self-selects (yes, that’s a schedule III drug) a 500mL antacid solution formulation containing 1:300 MgCO_3 . How many mEq of Mg^{2+} are there in RT’s bottle?

Molecular mass of $\text{MgCO}_3 = 84.3 \text{ g/mol}$

Cross-multiplication:

$$\frac{1 \text{ g}}{300 \text{ mL}} = \frac{x}{500 \text{ mL}} \quad x = 1.67 \text{ g}$$

$$\text{number of moles (n)} = \frac{1.67 \text{ g}}{84.3 \frac{\text{g}}{\text{mol}}} = 0.0198 \text{ mol} = 19.8 \text{ mmol}$$

Cross-multiplication:

$$\frac{\text{valence 1}}{19.8 \text{ mEq}} = \frac{\text{valence 2}}{x} \quad x = 39.6 \text{ mEq}$$

Problem Type 17 – Calculating osmolarity

Milliosmoles (mOsmols) are the number of millimoles of a substance multiplied by the number of ions or parts of the substance that can dissociate into when placed in a solution. These ions or parts affect the solution’s osmolarity. Knowing the number of mOsmol in a solution allows you to classify the osmolarity of that solution as hypotonic, hypertonic, or isotonic. A solution of 0.9 % NaCl is isotonic with the human body fluids. A hypertonic solution will cause cells to shrink, as water moves out of the cells to dilute the concentration of the hypertonic solution. Conversely, a hypotonic solution will cause cells to swell (and potentially lyse), as water moves into the cells (from a lower concentration of solutes in the hypotonic solution to a higher concentration of solutes in the cells).

Example 42:

How many mOsmol are there in 50mL of a 3% w/v $\text{Ca}_3(\text{PO}_4)_2$ solution that fully dissociates into its ions?

$\text{Ca}_3(\text{PO}_4)_2$ dissociates into 5 ions:

3 x Ca^{2+}

2 x PO_4^{3-}

Molecular mass of $\text{Ca}_3(\text{PO}_4)_2 = 310 \text{ g/mol} = 310\text{mg/mmol}$

Therefore, 1 mmol of $\text{Ca}_3(\text{PO}_4)_2 = 310\text{mg} = 5 \text{ mOsmol}$

The mass of $\text{Ca}_3(\text{PO}_4)_2$ in the specified solution:

$$50\text{mL} \times \frac{3\text{g}}{100\text{mL}} = 1.5\text{g} = 1500\text{mg}$$

Cross-multiplication:

$$\frac{1500 \text{ mg}}{x} = \frac{310 \text{ mg}}{5 \text{ mOsmol}} \quad x = 24.2 \text{ mOsmol}$$

Problem Type 18 – Determining the infusion rate of an intravenously-administered drug

Calculating the infusion rate is important when attempting to maintain a certain drug plasma concentration in a patient who is receiving intravenous therapy.

Example 43:

BZ is a 26-year-old male who was admitted to the hospital for a severe asthma attack. He was started on a theophylline infusion. Theophylline has a very narrow therapeutic window and so the pharmacist would like to maintain the steady state concentration at $40 \mu\text{mol/L}$ (drug plasma levels will be regularly monitored). Assuming that theophylline is 100% renally cleared (in reality it's actually < 15% renally cleared) and BZ has normal renal function, at what infusion rate (R_0) should theophylline be administered to BZ?

Mass = 100 kg

Height = 6 foot 4 inches

SCr = 100 $\mu\text{mol/L}$

$$\begin{aligned} \text{BZ's ideal body weight in kg (formula for males)} \\ &= 50 + 2.3 \times \text{number of inches above 5 feet in height} = 50 + 2.3 \times 16 \\ &= \mathbf{86.8 \text{ kg}} \end{aligned}$$

Estimate BZ's renal function:

The Cockcroft-Gault formula gives the result in units of mL/min

$$\begin{aligned} \text{CrCl} &= \frac{(140 - \text{age}) \times \text{ideal body weight}}{\text{serum creatinine} \left(\frac{\mu\text{mol}}{\text{L}} \right)} \times 1.2 \text{ (multiply by 0.85 if female)} \\ &= \frac{(140 - 26) \times 86.8}{100} \times 1.2 = \mathbf{118.7 \text{ mL/min}} \end{aligned}$$

$$C_{ss} = \frac{R_0}{\text{CrCl}}$$

Cl (clearance) happens to be 100% renal in this example, therefore substitute the estimated CrCl for Cl

$$R_0 = C_{ss} \times \text{CrCl} = 40 \frac{\mu\text{mol}}{\text{L}} \times 0.1187 \frac{\text{L}}{\text{min}} = \mathbf{4.75 \frac{\mu\text{mol}}{\text{min}}}$$

Problem Type 19 – Calculating the amount of ingested calories

Problems of this type have many forms including calculating the number of calories for patients who are trying to manage their weight or for patients who cannot tolerate a large calorie count secondary to organ dysfunction.

Example 44:

TK presents to the hospital with signs of acute pancreatitis secondary to alcohol abuse. After diagnostic tests, she is transferred to the inpatient ward and is to have nothing by mouth (NPO) for 10 days, thus she requires total parenteral nutrition.

She is given 2L of TPN daily, which is composed of:

Protein (4 cal/g) = 1:14

Dextrose (4 cal/g) = 8% w/v

Lipids (9 cal/g) = 2% w/v

Alcohol* (7 cal/g) = 1:100

*Note: TK prefers not to take any medications if possible, so rather than giving her haloperidol or benzodiazepines to treat any alcohol withdrawal symptoms, we add some alcohol to her TPN. Calculate the total number of daily calories TK receives from her TPN.

Quantity of calories obtained from each component:

Protein:

$$\frac{1 \text{ g}}{14 \text{ mL}} = \frac{x}{2000 \text{ mL}} \quad x = 142.9 \text{ g}$$

$$\text{Calories} = 142.9 \text{ g} \times 4 \text{ cal/g} = \mathbf{571.4 \text{ cal}}$$

Dextrose:

$$\frac{8 \text{ g}}{100 \text{ mL}} = \frac{x}{2000 \text{ mL}} \quad x = 160 \text{ g}$$

$$\text{Calories} = 160 \text{ g} \times 4 \text{ cal/g} = \mathbf{640 \text{ cal}}$$

Lipids:

$$\frac{2 \text{ g}}{100 \text{ mL}} = \frac{x}{2000 \text{ mL}} \quad x = 40 \text{ g}$$

$$\text{Calories} = 40 \text{ g} \times 9 \text{ cal/g} = \mathbf{360 \text{ cal}}$$

Alcohol:

$$\frac{1 \text{ g}}{100 \text{ mL}} = \frac{x}{2000 \text{ mL}} \quad x = 20 \text{ g}$$

$$\text{Calories} = 20 \text{ g} \times 7 \text{ cal/g} = \mathbf{140 \text{ cal}}$$

$$\Sigma \text{ Calories} = (571.4 + 640 + 360 + 140) \text{ calories} = \mathbf{1711.4 \text{ calories}}$$

Problem Type 20 – Calculating the amount or concentration of a stock solution required to make a solution of different strength or volume (for the same active ingredient)

This type of relation can be described by the equation $C_1V_1 = C_2V_2$. It is one of the most common types of calculations used in compounding. Knowing three of the variables allows us to calculate the fourth.

Example 45:

You have a stock solution of dextromethorphan syrup of concentration 20mg/mL. You wish to make 50mL of 2.5mg/mL dextromethorphan syrup for a patient. How much diluent do you need to make this solution?

$$C_1V_1 = C_2V_2$$

$$C_1 = 2.5 \text{ mg/mL}$$

$$C_2 = 20 \text{ mg/mL}$$

$$V_1 = 50\text{mL}$$

$$V_2 = ?$$

$$V_2 = C_1V_1 / C_2$$

$$V_2 = \frac{2.5 \frac{\text{mg}}{\text{mL}} \times 50 \text{ mL}}{20 \frac{\text{mg}}{\text{mL}}} = 6.25 \text{ mL}$$

$$V_{\text{diluent}} = V_1 - V_2 = 50\text{mL} - 6.25\text{mL} = \mathbf{43.75\text{mL}}$$

Problem Type 21 – Calculating quantities of ingredients for compounding a specified strength of a formulation, given different strengths of stock formulations using the alligation method

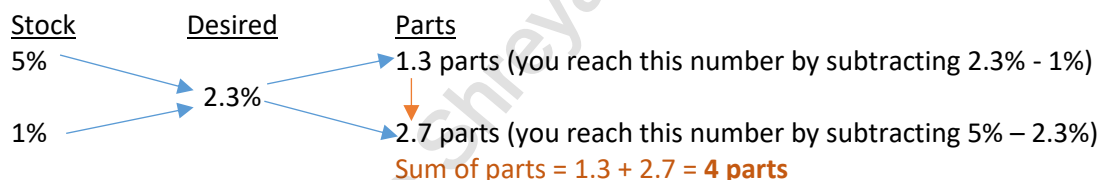
When you have 2 stock formulations and are asked to create your own formulation with a strength that lies in between those of the stock formulations, using the alligation method can help you determine the proportions of the stock formulations that you need to mix in order to achieve your desired strength.

Example 46:

a) EJ comes in with a prescription for 2.3% hydrocortisone ointment (Mitte: 60g). You have hydrocortisone 1% and 5% stock ointments available. How would you compound this ointment for EJ?

Use the alligation method

Cross-subtract using the following schematic:



For the 5% ointment:

$$\frac{1.3 \text{ parts}}{4 \text{ total parts}} = 32.5\%$$

$$32.5\% \text{ of } 60\text{g} = \mathbf{19.5\text{g}}$$

For the 1% ointment:

$$\frac{2.7 \text{ parts}}{4 \text{ total parts}} = 67.5\%$$

$$67.5\% \text{ of } 60\text{g} = \mathbf{40.5\text{g}}$$

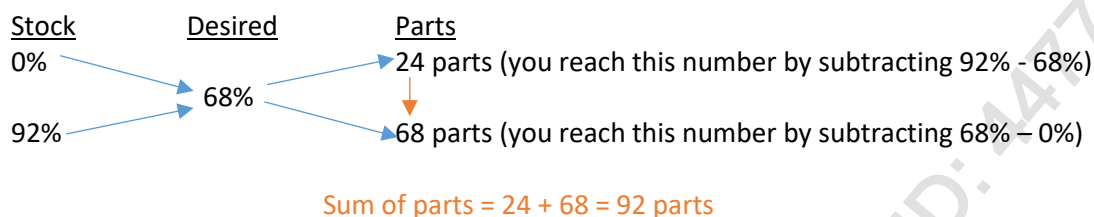
Therefore, to compound EJ's ointment, you would geometrically mix 19.5g of the 5% stock ointment with 40.5g of the 1% stock ointment in order to make 60g of 2.3% ointment.

b) EJ has an older prescription in the system to be dispensed as needed for athlete's foot. The prescription is for 68% of terbinafine cream in a corticosteroid base. You have 92% terbinafine cream as

stock formulation, with only 14g left, as the product seems to be on back-order. You have more than enough corticosteroid base in stock. What is the maximum quantity (mass) of 68% terbinafine cream that you can compound for EJ right now?

Use the alligation method.

Cross-subtract using the following schematic:



For the 0% terbinafine cream (the corticosteroid base):

$$\frac{24 \text{ parts}}{92 \text{ total parts}} = 0.26 = 26\%$$

For the 92% terbinafine cream:

$$\frac{68 \text{ parts}}{92 \text{ total parts}} = 0.74 = 74\%$$

We know that 74% of the mixture is composed of 92% terbinafine cream, of which we have only 14g. We would like to know how much cream in total we can compound. Cross-multiply:

$$\frac{14 \text{ g}}{74\%} = \frac{x}{100\%} \quad x = 18.9\text{g}$$

Therefore, we can make a maximum quantity of 18.9g of 68% terbinafine cream.

Problem type 22 – Compounding Calculations

Pharmaceutical compounding is the process in which pharmacists combine therapeutically active ingredients with pharmaceutical materials to prepare customized prescriptions and medication orders.

Example 47:

Using a vial containing 300,000 units of antibiotic G, how many milliliters of solvent should be added to the dry powder in preparing a solution that contains 14,000 units/mL?

Cross multiply:

$$\frac{14,000}{300,000} = \frac{1 \text{ mL}}{x \text{ mL}} \quad x = 21.43 \text{ mL}$$

Therefore, 21.43 mL of solvent should be used.

Example 48:

A pharmacist is filling a prescription and only has 2 mg unscored tablets of lorazepam.

Lorazepam 0.5 mg

Make # 90 tablets

(a) How many 2 mg tablets must be used?

(b) If the tablets from answer a are powdered and weigh a total of 425 mg, how many milligrams of the powder would provide the lorazepam required?

(c) If the desired total weight for the contents of each divided powder is 150 mg, how much lactose should be used as diluent?

(a) Calculate the total amount of lorazepam needed and how many tablets would then be required.

$$\begin{aligned}0.5 \text{ mg} \times 90 &= 45 \text{ mg} \\45 \text{ mg} \div 2 \text{ mg} &= \mathbf{22.5 \text{ tablets}}\end{aligned}$$

23 tablets are required (must use whole tablets so round up).

(b)

$$425 \times \frac{22.5}{23} = \mathbf{415.76 \text{ mg}}$$

415.76 mg of the powder would provide the lorazepam required.

(c)

$$\begin{aligned}150 \text{ mg} \times 90 &= 13,500 \text{ mg total weight} \\13,500 \text{ mg} - 415.76 \text{ mg} &= \mathbf{13,084.24 \text{ mg}}\end{aligned}$$

Therefore, a total of 13,084.24 mg of lactose is required.

Example 49:

You would like to compound some lidocaine suppositories for your dog who cannot stop scratching his behind secondary to a flea infection. A pure lanolin suppository has a mass of 1 g. The same size suppository containing 100 mg of lidocaine has a mass of 1.05 g.

- (i) What mass of lanolin do you need to make 14 suppositories of 100 mg lidocaine for your dog?
- (ii) What is the mass of lanolin displaced in each suppository?
- (iii) What is the density factor of lidocaine?

Answer:

- (i) We first look at a single suppository:

Mass of pure lanolin in a single pure lanolin suppository = 1 g

Mass of lanolin in a single combination suppository = 1.05 g

Mass of lidocaine in a single combination suppository = 100 mg = 0.1 g

Mass of lanolin in a single combination suppository = 1.05 g – 0.1 g = 0.95 g

Therefore mass of lanolin needed in 14 suppositories = 14 X 0.95 g = **13.3 g of lanolin**

$$\begin{aligned} \text{(ii)} \quad \text{Mass of lanolin displaced} &= \text{Lanolin in pure supp.} - \text{Lanolin in compounded sup.} \\ &= 1.0 \text{ g} - 0.95 \text{ g} \\ &= \mathbf{0.05 \text{ g}} \end{aligned}$$

$$\text{(iii)} \quad \text{Density factor for lidocaine} = \frac{\text{mass of lidocaine}}{\text{mass of lanolin displaced}} = \frac{0.1 \text{ g}}{0.05 \text{ g}} = \mathbf{2}$$

Example 50:

Determine the total quantities of each drug and lactose required to fill the following prescription using size 1 capsules (0.5 mL).

Drug A 30 mg

Drug B 65 mg

create 25 capsules

The capsules are filled individually and weighed, and the pharmacist found the following:

Capsule filled with drug A weighed 520 mg

Capsule filled with drug B weighed 370 mg

Capsule filled with lactose weighed 430 mg

First, find the diluent displacement weights for drug A and B

Drug A:

$$\frac{520 \text{ mg}}{430 \text{ mg}} = \frac{30 \text{ mg}}{x \text{ mg}} \quad \mathbf{x = 24.81 \text{ mg}}$$

Drug B:

$$\frac{370 \text{ mg}}{430 \text{ mg}} = \frac{65 \text{ mg}}{x \text{ mg}} \quad \mathbf{x = 75.54 \text{ mg}}$$

Next calculate the amount of diluent required per capsule:

$$430 \text{ mg} - (24.81 \text{ mg} + 75.54 \text{ mg}) = 329.65 \text{ mg lactose}$$

Finally, find the total quantity of each drug and diluent needed

$$\text{Drug A} = 30 \text{ mg} \times 25 = \mathbf{750 \text{ mg}}$$

$$\text{Drug B} = 65 \text{ mg} \times 25 = \mathbf{1625 \text{ mg}}$$

$$\text{Lactose} = 329.65 \text{ mg} \times 25 = \mathbf{8241.25 \text{ mg}}$$

Therefore, 750 mg of drug A, 1625 mg of drug B, and 8241.25 mg of lactose are required to fill all the capsules required.

Example 51:

Calculate the number of tablets containing the combination of spironolactone 15 mg and hydrochlorothiazide 15 mg that must be prepared using Hypromellose 1% as the suspension vehicle.

Spironolactone	10 mg/mL
Hydrochlorothiazide	10 mg/mL
Hypromellose 1%	140 mL

Amount of both drugs:

$$10 \frac{\text{mg}}{\text{mL}} \times 140 \text{ mL} = 1400 \text{ mg of spironolactone}$$

$$1400 \text{ mg} \div \frac{15 \text{ mg}}{\text{tablet}} = 93.3 \approx \mathbf{94 \text{ tablets}}$$

Therefore, 94 tablets are required.

Problem type 23 – Calculations involving plant extracts

Due to the ever-increasing interest in the use of alternative or complementary medicine, it is important to be able to understand calculations involving herbs and plant extracts. Crude drugs are the plant materials that are known to contain the active ingredient. Extracts are concentrated preparations of vegetable drugs and are commonly described as a multiple or percent extracts. Fluidextracts are liquid extractives where each mL is equivalent to 1g of crude drug. Botanic tinctures are alcoholic/hydroalcoholic solutions of plant extractives, and a potent tincture contains 1g of crude drug per 10 mL of tincture.

Example 52:

A batch of valerian is found to contain 15 mg in a 3-g sample. How many milligrams of valerian would be present in a 0.75 g dose of valerian from this batch?

Cross multiply:

$$\frac{3 \text{ g}}{0.75 \text{ g}} = \frac{15 \text{ mg}}{x \text{ mg}} \quad \mathbf{x = 3.75 \text{ mg}}$$

Example 53:

If there is 1 mg of active ingredient in each gram of a crude drug determine the concentration of active ingredient in (a) fluidextract, (b) 3x extract, (c) 600% extract, and (d) potent tincture.

(a) 1 mL of fluid extract is equivalent to 1 g of crude drug, then there would be 1 mg of active ingredient in 1 mL of fluid extract.

$$\frac{1 \text{ mg active ingredient}}{1 \text{ mL fluidextract}}$$

(b) A 3x extract represents, in each gram, 3 grams of crude drug.

Cross multiply:

$$\frac{1 \text{ g crude drug}}{3 \text{ g crude drug}} = \frac{1 \text{ mg active ingredient}}{x \text{ mg active ingredient}} \quad x = 3 \text{ mg active ingredient/gram extract}$$

(c) a 600% extract represents, in each gram, 6 grams of crude drug.

Cross multiply:

$$\frac{1 \text{ g crude drug}}{6 \text{ g crude drug}} = \frac{1 \text{ mg active ingredient}}{x \text{ mg active ingredient}} \quad x = 6 \text{ mg active ingredient/gram extract}$$

(d) A potent tincture represents in each 100 mL the active ingredient from 10g of crude drug. Therefore 0.1 grams would be required for 1 mL

Cross multiply:

$$\frac{1 \text{ g crude drug}}{0.1 \text{ g crude drug}} = \frac{1 \text{ mg active ingredient}}{x \text{ mg active ingredient}} \quad x = 0.1 \text{ mg active ingredient/mL tincture}$$

Example 54:

The herb black cohosh, when standardized as a powdered leaf, contains 0.4% of the agent triterpene glycosides. How many milligrams of triterpene glycosides is present in a capsule containing 75 mg of powdered black cohosh?

$$75 \times 0.4\% = \mathbf{0.30\ mg}$$

Therefore, 0.30 mg is contained in the capsule.

Problem type 24 – Calculation of active drug moiety

Drug substances are often created as salts or other complex chemical forms in order to facilitate solubility, biological absorption, and other characteristics. The active drug moiety of a drug compound can be calculated with this formula:

$$\frac{\text{Drug moiety} \left(\frac{g}{mole} \right)}{\text{Drug compound} \left(\frac{g}{mole} \right)} = \text{Drug moiety (fraction)}$$

Example 55:

The chemical formula for imipramine pamoate is $C_{19}H_{24}N_2 \cdot C_{23}H_{16}O_6$. Calculate (a) the molar mass of the base and salt forms, (b) the fraction of the imipramine base in the compound, (c) calculate the quantity of imipramine in a 200 mg dose of imipramine pamoate, and (d) calculate the quantity of imipramine pamoate needed to supply a 250 mg dose of imipramine.

(a) Molar mass of base:

$$(19 \times 12.01) + (24 \times 1.00) + (2 \times 14.01) = \mathbf{280.21 \frac{g}{mole}}$$

Molar mass of salt form:

$$280.21 + (23 \times 12.01) + (16 \times 1.00) + (6 \times 16) = \mathbf{668.44 \frac{g}{mole}}$$

(b)

$$\frac{280.21}{668.44} = \mathbf{0.42}$$

(c)

$$200\ mg \times 0.42 = \mathbf{84\ mg}$$

(d)

$$\frac{250 \text{ mg} \times 668.44 \frac{\text{g}}{\text{mole}}}{280.21 \frac{\text{g}}{\text{mole}}} = 596.37 \text{ mg}$$

Problem type 25 – Radiopharmaceutical calculations

The exponential decay law is used to calculate the rate of decay of unstable isotopes. The exponential decay law is expressed by the following formula:

$$N = N_0 e^{-lt}$$

Where N_0 is the amount of activity when time is zero and l is the half-life coefficient or disintegration constant.

The half-life of a radioisotope is the time it takes for the total number of atoms to be reduced in half. The half-life equation is:

$$T_{\frac{1}{2}} = \frac{0.693}{l}$$

It is important to note that radioactivity is measured in the SI unit becquerel (Bq) and historically has been measured in the curie (Ci). The conversion factor is:

$$1 \text{ Ci} = 3.7 \times 10^{10} \text{ Bq}$$

Example 56:

The original activity of a radioisotope is 20.5 MBq. After 24 days, if the activity is 3.75 MBq, what is (a) the disintegration constant, and (b) the half life?

(a) Convert the exponential decay law formula to its logarithmic form and then substitute.

$$\ln \frac{N}{N_0} = -lt$$

$$l = \frac{\ln \frac{N}{N_0}}{-t} = \frac{\ln \frac{3.75}{20.5}}{-24} = 0.071 \text{ days}^{-1}$$

(b) Substitute the values into the half-life formula

$$T_{\frac{1}{2}} = \frac{0.693}{0.071} = 9.76 \text{ days}$$

Therefore the disintegration constant is 0.071 days^{-1} and the half-life is 9.76 days.

Example 57:

A sample of radioactive iodine has an initial activity of 2.32 MBq. Its half life is 8.08 days. What is its activity at the end of exactly 27 days?

First, find the disintegration constant by rearranging the half life formula

$$l = \frac{0.693}{8.08 \text{ days}} = 0.086 \text{ days}^{-1}$$

Substitute values into the exponential decay formula

$$N = 2.32 \text{ MBq} \times e^{-27 \text{ days} \times 0.086 \text{ days}^{-1}} = 0.23 \text{ MBq}$$

Problem type 26 – Bioavailability Calculations

Bioavailability (F) refers to the proportion of the drug that enters systemic circulation when is introduced into the body and is able to have an active effect. Bioavailability ranges between 0 (no absorption) to 1 (complete absorption such as intravenous infusion). Absolute bioavailability refers to the comparison of the amount of drug absorbed from a dosage form to the amount delivered by an intravenous dose. Relative bioavailability refers to the comparison of amounts absorbed between different dosage forms or routes.

Pharmacists may also need to calculate the equivalent doses for two bioequivalent products using the following formula:

$$F_1 \times \text{Dose}_1 = F_2 \times \text{Dose}_2$$

Bioavailability can also be calculated using AUC data from a blood level-time curve using the following formula:

$$F = \frac{AUC_{\text{dosage form}}}{AUC_{\text{intravenous}}}$$

Example 58:

The oral bioavailability of 20-mg of drug Z is stated to be 55%. Concomitant administration with coffee reduces the bioavailability by approximately 30%. Calculate the quantity of drug Z following a 20-mg dose with (a) water and (b) coffee.

(a)

$$20 \text{ mg} \times 0.55 = \mathbf{11 \text{ mg}}$$

(b)

$$11 \text{ mg} \times (1 - 0.3) = \mathbf{7.7 \text{ mg}}$$

Therefore, the quantity of a 20-mg drug Z after oral administration with water is 11 mg and with coffee is 7.76 mg.

Example 59:

If the bioavailability factor for 250-mg tablet of a drug is 0.8 compared to the bioavailability factor of 1.0 for an injection of the same drug, how many milliliters of the injection containing 40 mg/mL would be considered bioequivalent to the tablet?

Substitute the values into the bioequivalence formula:

$$\begin{aligned} 0.8 \times 250 \text{ mg} &= 1 \times \text{Dose}_{\text{injection}} \\ \text{Dose}_{\text{injection}} &= 200 \text{ mg} \\ \text{Dose}_{\text{injection}} \text{ in mL} &= 200 \text{ mg} \div \frac{40 \text{ mg}}{\text{mL}} = \mathbf{5 \text{ mL}} \end{aligned}$$

Therefore, 5 mL of the injection would be considered bioequivalent.

Example 60:

If the AUC for an oral dose of a drug administered by tablet is 3.86 mcg h/mL and the intravenous dose is 12.32 mcg h/mL, calculate the bioavailability of the oral dose of the drug.

$$F = \frac{AUC_{\text{oral}}}{AUC_{\text{intravenous}}} = \frac{3.86 \text{ mcg h/mL}}{12.32 \text{ mcg h/mL}} = \mathbf{0.31 \text{ or } 31\%}$$

Therefore, the bioavailability of the oral dose is 0.31.

Problem type 27 – Volume of distribution Calculations

The volume of distribution of a drug is the hypothetical volume required to contain the total amount of administered drug at the same concentration that is observed in the blood plasma.

$$Vd = \frac{D}{C_p}$$

Where D is the total amount of drug in the body and C_p is the drug's plasma concentration.

Example 61:

A patient is receiving a single intravenous dose of 400 mg of a drug substance that produced an immediate blood concentration of 22.56 mg of drug per milliliter. Calculate the volume of distribution.

$$Vd = \frac{400 \text{ mg}}{22.56 \text{ mg/mL}} = 17.73 \text{ mL}$$

Example 62:

After 5 hours following administration of a drug, a patient weighing 96 kg was found to have a drug blood level of 45 mg/L. Assuming the volume of distribution is 23% of body weight, calculate the total amount of drug in the body 5 hours after administration.

$$Vd = 96 \times .23 = 22.08 \text{ L}$$

$$D = Vd \times C_p = 22.08 \times 45 \frac{\text{mg}}{\text{L}} = 993.6 \text{ mg}$$

Problem type 28 – Drug elimination Calculations

Pharmacists may be required to calculate rates of drug elimination. Most drug substances follow first order kinetics and are related to this formula:

$$C_t = C_0 e^{-K_{el}t}$$

Where C_t is the drug at volume t , C_0 is the drug given intravenously, and K_{el} is the elimination rate constant.

The elimination rate constant of a drug following first order kinetics may be calculated using this formula:

$$K_{el} = \frac{\ln 2}{t_{1/2}}$$

Example 63:

(a) Calculate the elimination rate constant for a drug that has a half life of 30 minutes? (b) If a patient is initially given a bolus dose of 480 mg, how much would be left in the blood after 175 minutes?

(a)

$$K_{el} = \frac{\ln 2}{30 \text{ min}} = 0.023 \text{ min}^{-1}$$

Therefore, the elimination rate constant is 0.023 min^{-1}

(b)

$$C_t = 480 \text{ mg} \times e^{-0.023 \text{ min}^{-1} \times 175 \text{ min}} = \mathbf{8.57 \text{ mg}}$$

Therefore, there is 8.57 mg of the drug left after 175 minutes.

Problem type 29 – Calculating Drug Therapy Costs

In drug therapy, the cost differential between different drug products and treatments can be a potential consideration.

Example 64:

A β -blocker is available in a three-times-a-day tablet at \$22.20/80 tablets, in a twice-a-day tablet at \$31.00/80 tablets, and in a one-a-day at \$56.45/40 tablets. Which form would be the most economical for a patient and what is the cost for 80 days of treatment?

Calculate the costs of all available treatments:

$$\text{three times a day} = \frac{\$22.20}{80 \text{ tabs}} \times 3 \text{ times a day} \times 80 \text{ days} = \mathbf{\$66.60}$$

$$\text{two times a day} = \frac{\$31.00}{80 \text{ tabs}} \times 2 \text{ times a day} \times 80 \text{ days} = \mathbf{\$62.00}$$

$$\text{one time a day} = \frac{\$56.45}{40 \text{ tabs}} \times 80 \text{ days} = \mathbf{\$112.90}$$

The twice-a-day tablet regimen would be the most cost-effective treatment.

Example 65:

A physician prescribes a patient thirty 50-mg tablets of drug Z and instructs the patient to split the tablets for a 60-day supply. One hundred 50-mg tablets costs \$13.25 and one-hundred 25-mg tablets costs \$11.75. Calculate the patient's savings on a 60-day supply.

$$50 \text{ mg tablets} = 30 \text{ tablets} \times \frac{13.25}{100} = \$3.98$$

$$25 \text{ mg tablets} = 60 \text{ tablets} \times \frac{11.75}{100} = \$7.05$$

$$\text{Total savings} = \$7.05 - \$3.98 = \$3.07$$

Therefore, the patient saves \$3.07 on a 60-day supply.

Problem Type 30 – Absolute Risk Reduction, Number Needed to Treat, Relative Risk, Relative Risk Reduction

Clinical trials express their results using a variety of patient-health indicators such as risk reduction, number of patients needed to treat, and odds ratios. The risk reduction refers to how much a patient's risk of a certain disease or conditions is decreased (ideally) when the patient undergoes a certain therapy. The number needed to treat refers to how many patients need to be treated with the therapy in question in order for just one of the patients (on average of course) in the group to derive a benefit. The odds ratio compares a patient's chances of developing a disease/condition when the patient is taking the treatment in the experiment, versus when the patient is not taking that treatment (they could be taking placebo or an alternative standard therapy).

Absolute Risk Reduction (ARR)

$$\text{ARR} = \text{proportion of events in control group} - \text{proportion of events in experimental group}$$

$$\text{ARR} = \text{CER} - \text{EER}$$

Number Needed to Treat (NNT)

$$\text{NNT} = \frac{1}{\text{ARR}}$$

Relative Risk (RR)

$$\text{RR} = \frac{\text{proportion of events in experimental group}}{\text{proportion of events in control group}}$$

Relative Risk Reduction (RRR)

$$\text{RRR} = 1 - \text{RR}$$

Odds Ratio (OR)

An odds ratio compares the likelihood of developing a disease if the experimental treatment was given to the likelihood of developing the same disease if the experimental treatment was NOT given. If the experimental treatment is effective (ideally), the odds ratio would be < 1 , meaning that less patients who took the treatment developed the disease as compared to those patients who did NOT take the treatment.

	Developed the disease	Did NOT develop the disease
Took the experimental treatment	A	B
Did NOT take experimental treatment	C	D

$$OR = \frac{\frac{A}{B}}{\frac{C}{D}} = \frac{A \times D}{B \times C}$$

Sensitivity and Specificity

Sensitivity refers to the likelihood of having a true positive result when you have the disease

Specificity refers to the likelihood of having a true negative result when you do NOT have the disease

	Has the disease	Does NOT have the disease
Positive test	A	B
Negative test	C	D

A = True positives

B = False positives

C = False negatives

D = True negatives

$$\text{Sensitivity} = \frac{A}{A + C}$$

$$\text{Specificity} = \frac{D}{B + D}$$

SpIN: If the test or categorization method has a high specificity, you can rule the result in (meaning that it's likely a true positive)

SnOUT: If the test or categorization method has a high sensitivity, you can rule the result out (meaning that it's likely a true negative)

References:

Barratt A et al. *Tips for learners of evidence-based medicine: Relative risk reduction, absolute risk reduction and number needed to treat*. CMAJ. 2004;171(4). Accessed April 2014.
<http://www.cmaj.ca/content/171/4/353.full.pdf>.

Example 66:

A new herb is being investigated for its potential use in the management of insomnia. Patients were given either placebo or the herb and the following results were observed:

	Insomnia	No insomnia
Taking the herb	32	70
Taking placebo	20	43

Calculate the RRR, ARR, and OR.

$$RR = \frac{\text{proportion of events in experimental group}}{\text{proportion of events in control group}} = \frac{\frac{32}{70+32}}{\frac{20}{20+43}} = \frac{0.314}{0.317} = \mathbf{0.99}$$

$$RRR = 1 - RR = 1 - 0.99 = \mathbf{0.01}$$

$$ARR = |\text{proportion of events in control group} - \text{proportion of events in experimental group}| \\ = |0.317 - 0.314| = \mathbf{0.003}$$

$$OR = \frac{32 \times 43}{20 \times 70} = \frac{1376}{1400} = \mathbf{0.98}$$

This new herb does not seem to have significant anti-insomnia effects. If we wanted to see how many patients would need to be treated so that one of them could benefit, we could calculate the NNT.

$$NNT = \frac{1}{ARR} = \frac{1}{0.003} = \mathbf{333 \text{ patients}}$$

Example 67:

A new tool has been developed for screening fungal vaginal infections. The hope is that if the patient tests positive, she can skip the visit to the doctor and obtain non-prescription medication from a pharmacy (ex. fluconazole).

	Thrush present	No thrush present
Positive test	25	0
Negative test	8	39

Calculate the sensitivity and specificity of this new tool. Is this tool likely to lead to significant unnecessary medicating?

$$\text{Sensitivity} = \frac{\text{true positives}}{\text{true positives} + \text{false negatives}} = \frac{25}{25 + 8} = \mathbf{0.76}$$

$$\text{Specificity} = \frac{\text{true negatives}}{\text{true negatives} + \text{false positives}} = \frac{39}{39 + 0} = \mathbf{1}$$

No, this tool is not likely to lead to significant unnecessary medicating, as it has a high specificity. A high specificity means that if the patient tests positive, they are likely to actually have the fungal infection, in which case, treatment is warranted.

Problem Type 31 - Determining the minimum weighable quantity (MWQ) of a balance

Regardless of the type of pharmacy setting you find yourself in, you may be asked to compound (or weigh out) certain medications.

Depending on the sensitivity of the balance available, you can determine the lowest amount of

substance that can be accurately measured.

The common acceptable standard used in the pharmacy industry is that weight measurements must be within 5% of the true amount. Every scale has something called a Minimum Weighable Quantity (MWQ). This is smallest mass you can weigh and still be within 5% of the true weight. If you try to weigh less than the MWQ, your error could be greater than 5% (which is considered unacceptable in pharmacy practice).

But quite often you are not given the Minimum Weighable Quantity. Instead you are given the maximum (i.e. 5%) error when weighing the MWQ which is called the **sensitivity**. The relationship between the MWQ and sensitivity is then:

$$\text{Maximum Error when weighing the MWQ} = 5\% \times \text{MWQ} = \text{sensitivity}$$

This is equivalent to writing

$$\text{MWQ} = \frac{\text{sensitivity}}{5\%}$$

Example 68:

Your lab has recently purchased a new balance with a sensitivity of 0.9 mg. Calculate the minimum weighable quantity (MWQ) of the balance.

$$\text{MWQ} = \frac{\text{sensitivity}}{5\%} = \frac{0.9 \text{ mg}}{0.05} = \mathbf{18 \text{ mg}}$$

Therefore, the minimum amount you can reliably weight using this balance is 18mg. By “reliably” we mean that whatever mass the scale reports is guaranteed to have at most 5% error.

Problem Type 32 – Calculating percentage of error

Measurements in community pharmacy are never absolutely accurate and its important to recognize limitations and errors. Percentage of error may be defines as the maximum potential error multiplied by 100 and divided by the quantity desired:

$$\text{Percentage of error} = \frac{\text{Error} \times 100\%}{\text{Quantity desired}}$$

Example 69:

Using a graduated cylinder, a pharmacist measured 50 mL of liquid. Subsequent examination it was determined that the pharmacist had actually measured 53 mL. What was the percentage of error?

$$\text{Volume of error} = 53 \text{ mL} - 50 \text{ mL} = \mathbf{3 \text{ mL}}$$

$$\frac{3 \text{ mL} \times 100\%}{50 \text{ mL}} = \mathbf{6\%}$$

Problem Type 33 – Chemotherapy Dosage regimens

Chemotherapy treatment plans commonly include two-agent, three-agent, or four-agent regimens. The days of treatment follow a prescribed format of written instructions, with D for day followed by the days

of treatment during the cycle. A dash means 'to' and a comma is 'and', so D 1-5 means days 1 to 5 and D 1,5 means day 1 and 5.

Example 70:

A patient measuring 180 cm in height and weighing 74 kg is on the following chemotherapy regimen:

Cycle: 28 days; repeat for 3 cycles

Vinorelbine, 30 mg/m², IV, D 1,8,15

Cisplatin, 80 mg/m², IV, D 1-5

Calculate the total quantity of each drug that is administered during the regimen, after all 3 cycles.

First, calculate the patients BSA:

$$BSA = \sqrt{\frac{180 \text{ cm} \times 74 \text{ kg}}{3600}} = 1.92 \text{ m}^2$$

Next calculate the total dosage of both drugs after 3 cycles.

$$\text{Vinorelbine} = 30 \frac{\text{mg}}{\text{m}^2} \times 1.92 \text{ m}^2 (\text{BSA}) \times 3 (\text{days}) \times 3 (\text{cycles}) = 518.4 \text{ mg}$$

$$\text{Cisplatin} = 80 \frac{\text{mg}}{\text{m}^2} \times 1.92 \text{ m}^2 (\text{BSA}) \times 5 (\text{days}) \times 3 (\text{cycles}) = 2304 \text{ mg}$$

Therefore, the total dosage of vinorelbine is 518.4 mg and for cisplatin is 2304 mg.

Practice Problems

Question 1 – The following question is based on the following patient information:

Name: Candy

Sex: Female

Age: 29

Height: 165 cm

Weight: 65 kg

Serum Creatinine: 100 $\mu\text{mol/L}$

Candy is diagnosed with bladder cancer and she is started on the following Chemotherapy regimen:

Cycle: 28 days; repeat for 5 cycles

Methotrexate, 30 mg/m^2 , D 1, 15, and 22

Vinblastine, 3 mg/m^2 , D 2, 15, and 22

Doxorubicin, 30 mg/m^2 , D 2

Cisplatin, 70 mg/m^2 , D 2

- Calculate the total amount of each medication that is administered per cycle.
- What is the intravenous infusion rate if the total amount of doxorubicin is delivered in 4 hours from a 1L solution?
- Calculate the steady state concentration of doxorubicin, assuming it is 100% renally cleared.
- The pK_a of methotrexate is 4.8. The ratio of ionized to unionized methotrexate in the IV solution is 75:1. What is the pH of the methotrexate IV solution.
- If an injection of cisplatin containing 8 mg/mL is available, what is the volume of injection required per cycle to treat the patient.

Question 2 – A potential new herb is being investigated to manage narcolepsy. 150 patients were recruited with 80 being given herb and 70 given a placebo. 60% of the patients given the herb and 30% given the placebo still had narcolepsy.

- Calculate the RRR, ARR, OR, and NNT.
- If there is 2 mg of active drug in each gram of crude drug, determine the concentration of active ingredient in:
 - Fluidextract
 - 400% extract
 - Potent tincture

Question 3 – DK has a history of anemia and takes iron sulfate supplements. Iron sulfate is commonly administered as $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$ which dissociates to 80% in concentration.

- (a) How many milliliters of isotonic solution can be produced from 1 g of FeSO_4 ?
- (b) How much 0.9% w/v sodium chloride is required to produce a solution having a 1% w/v concentration of FeSO_4 ?
- (c) Calculate the milliequivalents of Fe^{2+} in 1 gram of FeSO_4 .
- (d) If DK is taking 10 mL of the isotonic solution per day, what mass of Fe^{2+} is found in that dose?
- (e) What fraction of active drug moiety is found in $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$?
- (f) How much $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$ is required to supply a 300 mg dose of iron (II)?

Question 4 – While working as a community pharmacy you receive an order of 80 tablets of 60-mg phenobarbital. However, you currently only have 80-mg tablets available.

- (a) How many 80-mg tablets are needed to make 80 tablets of 60-mg tablets?
- (b) If the tablets from part (a) are powdered and weight a total of 5650 mg, how many milligrams of the powder would provide the phenobarbital required?
- (c) If the desired weight of each tablet is 120 mg, how much lactose is required as diluent?
- (d) If the bioavailability of the 60-mg phenobarbital tablets is 0.44 compared to the 1.0 for a phenobarbital injection, how many mL of an injection containing 30 mg/mL phenobarbital would be bioequivalent to one tablet?
- (e) The patient is concerned about the price of the medication. If the cost of 60-mg tablets is \$15.47/100 tablets and \$18.76/100 tablets for the 80-mg tablets, should the patient wait for the 60-mg tablets to become available? Disregard all other potential cost factors other than medication price.

Question 5 – Scientists have developed a new test for a bloodborne disease that uses substance T. The following results were found:

	Disease present	No disease present
Positive test	30	5
Negative test	12	43

- (a) What is the sensitivity of this test?
- (b) What is the specificity of this test?
- (c) The scientists in question purchased a balance with a sensitivity of 0.3 mg. What is the MWQ?
- (d) A scientist measured out 35 mg of Substance X. However, subsequent examination found that 41 mg was actually measured out. What is the percentage of error?

Question 6 – The following question is based on the following patient information.

Name: Lucas

Sex: Male

Height: 185 cm

Mass: 95 kg

Diagnosis: Bipolar disorder

Medication List:

Valproic Acid 25 mg/kg/day

Escitalopram 20 mg/day

Risperidone 6 mg/day

Alprazolam 3 mg as needed

- (a) What is the daily dose of valproic acid?
- (b) In a given month, Lucas uses alprazolam between 12 – 18 times on average. What is the maximum and minimum dosage per month?
- (c) The oral bioavailability of 20-mg escitalopram is 0.67 when administered alone. This is decreased by 15% when administered with valproic acid. Calculate the quantity of escitalopram when administered with valproic acid.
- (d) What is the elimination constant of risperidone if the half life is 3 hours?
- (e) Using the elimination constant found in (d), how much risperidone would be left in the blood after 75 minutes?
- (f) Using the dosage found in part (a), what is valproic acid's volume of distribution if the blood concentration is 123 mg/mL immediately after administration?

Question 7 – A formula for a compounded ophthalmic solution is shown below:

Tobramycin sulfate 250 mg

Diclofenac sodium 75 mg

Ketotifen fumarate 25 mg

Sodium Chloride 800 mg

Sterile Water 100 mL

- (a) Tobramycin sulfate is a 7-ion electrolyte and has a molecular weight of 1425.45. It dissociates 80% at a certain concentration, calculate its dissociation factor and E-value.
- (b) The potency of ketotifen fumarate is 564 to 723 mcg of ketotifen activity per milligram. What is the range of ketotifen activity in this formulation?
- (c) Diclofenac sodium is a 2-ion electrolyte with a dissociation factor of 1.8 and a molecular weight of 318.13. Calculate the E-factor of diclofenac sodium.
- (d) Ketotifen fumarate is a 2-ion electrolyte with a dissociation factor of 1.8 and a molecular weight of 424.498 g/mol. Calculate the E-factor of ketotifen fumarate.
- (e) Is the quantity of sodium chloride enough to create an isotonic solution?
- (f) How much of each ingredient would be required to create 30 mL of solution?

Question 8 – A pharmacist prepares a 1 L buffer solution containing 0.2 M concentration of sodium acetate and 0.25 M of acetic acid. The pK_a value of acetic acid is 4.76 at 25°C.

- (a) Calculate the pH and pOH of the solution.
- (b) What is the molar ratio of salt/acid in this buffer solution?
- (c) What is the change in pH after adding 0.05 mol of sodium hydroxide to the buffer solution?
- (d) What is the change in pH after adding 0.1 mol of hydrochloric acid?
- (e) What is the pK_b value of sodium acetate?

Question 9 – A pharmacist is producing an oral suspension from oseltamivir capsules and the following parameters must be followed:

- The suspension must have a drug concentration of 7 mg/mL for an adult
 - The volume of compound is based on the patient's weight:
 - ≤ 33 kg = 30 mL
 - 34 – 51 kg = 40 mL
 - ≥ 52 kg = 50 mL
 - Cherry syrup is used as the vehicle
 - 50-mg capsules are to be used
- (a) How many 50-mg capsules must be used to compound the suspension for a 50 kg adult?
 - (b) If the prophylactic dose for 50 kg adult is 60 mg once daily, how many millilitres of the compounded suspension would constitute a dose?
 - (c) The pharmacist wishes to convert the drug dose for a pediatric patient using Young's rule and Clark's rule. Calculate the pediatric drug concentration using both methods for an 8-year-old.
 - (d) Based on the answer from (c), calculate the appropriate number of 50-mg capsules that must be used to compound a suspension for a 30-kg 8-year old.

Question 10 – A concentrated acid consists of 60% w/w acid and has a specific gravity of 1.56.

- (a) How many milliliters of the acid would be required to prepare 55 mL of a 10% w/v solution?
- (b) If the strength of the acid is mistakenly read as 60% w/v, how much of the concentrated acid would be used to prepare the solution in (a)?
- (c) What would be the percent error in the amount of concentrated glycolic acid measured in (b)?
- (d) What would be the resulting percent strength of the diluted acid solution in (b) due to the mistake?

Question 11 – A 89 kg cardiology patient receives an initial heparin bolus dose of 60 units/kg followed by a heparin drip at 15 units/kg/h. The heparin concentration was 5000 units per 100 mL and the intravenous set delivered 20 drop per milliliter.

- (a) Calculate the initial heparin bolus dose, in both units and milliliters, if the product administered contained 1000 units/mL.
- (b) The patient is found to be underdosed and their heparin rate should be increased by 40%. Calculate the revised dosage in units per kilogram per hour.
- (c) Calculate the revised flow rate using the answer from (b).

Question 12 – The following is a formula for a testosterone nasal spray:

Testosterone	1.5 g
Alcohol	20 mL
Propylene glycol	25 mL
Benzalkonium chloride	10 mg
Purified water	100 mL

- (a) Calculate the quantity of each ingredient required to fill twenty-four 25 mL nasal spray bottles of the formula.
- (b) Benzalkonium chloride is available as a 1:250 w/v stock solution. How many milliliters would provide the amount determined in (a).
- (c) If the propylene glycol is found to be contaminated with 2.3 ppm of a foreign substance would be contained in each bottle of nasal spray?
- (d) If the pharmacist weighed the testosterone using a sensitive electronic balance and found 10.36 g was weighed rather than the calculated quantity in (a), what is the percent error?
- (e) If the pharmacist decides to use testosterone cypionate injection, 300 mg/mL, as a source of testosterone, calculate the quantity needed for the amount in (a) if the molecular weight of testosterone cypionate is 412.6 g/mol and testosterone is 288.4 g/mol.
- (f) The normal blood level of testosterone in males is between 270 and 1070 ng/dL. If a 10 mL blood sample is found to contain 53.9 ng of testosterone would the patient's testosterone level fall within the normal range?

Question 13 – A nasal spray contains 0.2% oxymetazoline hydrochloride and 75 mg/mL of benzalkonium chloride as a preservative. A container is capable of delivering 250 metered sprays of 0.162 mL each.

- (a) Calculate the quantity of oxymetazoline hydrochloride in each spray.
- (b) Calculate the percentage strength and ratio strength of benzalkonium chloride in the preparation.

- (c) If the nasal spray comes in 45 mL bottles, how many milliliters are left after 225 metered sprays?
- (d) The recommended dose of the spray to treat allergies is one spray in each nostril twice daily. At this dose, how long will a 45 mL bottle last a patient?

Question 14 – A medication order calls for a patient to receive 0.2 mCi of sodium iodide for a thyroid function test. Sodium iodide is available in 3.7-MBq capsules and can be used up to 30 hours after measurement using a radioactivity calibration system.

- (a) How many capsules should be dispensed to provide the prescribed dose? Round your answer to the nearest whole number.
- (b) How many mCi of radioactivity will be available at the 30-hour cut-off time if the half-life of sodium iodide is 13.2 hours?
- (c) A different patient was given an unknown dose of sodium iodide. After 18 hours, the radioactivity was measured and found to be 0.09 mCi. What was the original dose?

Question 15 – An order for an IV admixture is as follows:

Tramadol hydrochloride 30 mEq in 1 L of 5% dextrose

- (a) How many milliliters of a tramadol hydrochloride 10% w/v injection should be used to prepare this IV admixture?
- (b) What would be the osmolarity of the IV admixture solution? (Assume volumes are additive and complete dissociation)
- (c) If the flow rate of this solution is 30 mL/h, how many milliequivalents of tramadol would the patient receive daily?

Practice Problems Solutions

Question 1 Solution:

(a) First, calculate the patients BSA:

$$BSA = \sqrt{\frac{165 \text{ cm} \times 65 \text{ Kg}}{3600}} = 1.73 \text{ m}^2$$

Next, calculate the total amount of each drug

$$\text{Methotrexate} = 30 \frac{\text{mg}}{\text{m}^2} \times 1.73 \text{ m}^2 \times 3 (\text{days}) = 155.7 \text{ mg}$$

$$\text{Vinblastine} = 3 \frac{\text{mg}}{\text{m}^2} \times 1.73 \text{ m}^2 \times 3 (\text{days}) = 15.57 \text{ mg}$$

$$\text{Doxorubicin} = 30 \frac{\text{mg}}{\text{m}^2} \times 1.73 \text{ m}^2 = 51.9 \text{ mg}$$

$$\text{Cisplatin} = 70 \frac{\text{mg}}{\text{m}^2} \times 1.73 \text{ m}^2 = 121.1 \text{ mg}$$

(b) Infusion rate (R_0) = $\frac{51.9 \text{ mg}}{4 \text{ hours}} = 12.975 \frac{\text{mg}}{\text{hour}}$

(c) First calculate the patients ideal body weight in Kg

$$45.5 \text{ kg} + 2.3 \text{ kg} \times 5 = 57 \text{ Kg}$$

Next, estimate her creatinine clearance:

$$\begin{aligned} CrCl &= \frac{1.2 \times (140 - \text{age}) \times \text{ideal body weight}}{\text{serum creatinine}} \times 0.85 \\ &= \frac{1.2 \times (140 - 29) \times 57 \text{ Kg}}{100 \frac{\mu\text{mol}}{\text{L}}} \times 0.85 = 64.53 \frac{\text{mL}}{\text{min}} \times \frac{60 \text{ min}}{\text{hour}} = 3872.12 \frac{\text{mL}}{\text{hour}} \end{aligned}$$

Substitute values to solve for steady state concentration:

$$C_{ss} = \frac{R_0}{CrCl} = \frac{12.975 \text{ mg/hr}}{3872.12 \text{ mL/hr}} = 3.35 \frac{\mu\text{g}}{\text{mL}}$$

(d) HA is un-ionized and A^- refers to the base so the ratio of HA to A^- = $1/75 = 0.0133$

$$pH = pKa - \log \frac{HA}{A^-} = 4.8 - \log 0.0133 = 6.68$$

(e) $\text{Volume required} = 121.1 \text{ mg} \div \frac{8 \text{ mg}}{\text{mL}} = \mathbf{15.14 \text{ mL}}$

Question 2 Solution:

	Narcolepsy	No Narcolepsy
Taking the herb	48	32
Taking placebo	21	49

(a) Calculate RR, RRR, ARR, OR, and NNT

$$RR = \frac{\text{proportion of events in experimental group}}{\text{proportion of events in control group}} = \frac{\frac{48}{32+48}}{\frac{21}{21+49}} = \frac{0.6}{0.3} = \mathbf{2}$$

$$RRR = 1 - RR = 1 - 2 = \mathbf{-1}$$

$$ARR = |0.6 - 0.3| = \mathbf{0.30}$$

$$OR = \frac{48 \times 49}{21 \times 32} = \mathbf{3.5}$$

$$NNT = \frac{1}{ARR} = \frac{1}{0.3} = 3.3 \approx \mathbf{4 \text{ patients}}$$

(b) Calculate the needed value when there are 2 mg of active ingredient per gram of crude drug

i) Fluid Extract

$$\text{Fluid Extract} = \frac{\mathbf{2 \text{ mg of active ingredient}}}{\mathbf{1 \text{ mL fluidextract}}}$$

ii) A 400% extract represents, in each gram, 4 grams of crude drug

$$\frac{1 \text{ g of crude drug}}{4 \text{ g of crude drug}} = \frac{2 \text{ mg active ingredient}}{x \text{ mg active ingredient}} \quad x = \mathbf{8 \text{ mg} \frac{\text{active ingredient}}{\text{gram extract}}}$$

iii) A potent tincture

$$\frac{1 \text{ g crude drug}}{0.1 \text{ g crude drug}} = \frac{2 \text{ mg active ingredient}}{x \text{ mg active ingredient}} \quad x = \mathbf{0.2 \frac{\text{mg active ingredient}}{\text{mL tincture}}}$$

Question 3 Solution:

- (a) First calculate the molecular weight (MW) of FeSO_4 = MW of Fe + MW of S + 4 X (MW of O)
 $= 55.845 + 32.065 + 15.999 \times 4 = 151.91$ atomic mass units (when rounded to 2 digits)

FeSO_4 is a 2-ion electrolyte and given the 80% dissociation, 100 particles of FeSO_4 yields:

80 Fe^{2+} ions

80 SO_4^{2-} ions

20 undissociated particles

Total: 180 particles

This leads to an i-factor of $180/100 = 1.8$

Next we calculate the E-value of FeSO_4 as follows:

$$\begin{aligned} E \text{ value of } \text{FeSO}_4 &= \frac{\text{Molecular Weight of NaCl}}{i \text{ Factor of NaCl}} \cdot \frac{i \text{ factor of } \text{FeSO}_4}{\text{Molecular Weight } \text{FeSO}_4} \\ &= \frac{58.5}{1.8} \times \frac{1.8}{151.91} = 0.39 \end{aligned}$$

Now calculate the volume of isotonic solution that can be produced from 1 g of FeSO_4 :

$$\begin{aligned} \frac{0.39 \text{ g}}{x \text{ mL}} &= \frac{0.9 \text{ g}}{100 \text{ mL}} \\ x &= \mathbf{43.33 \text{ mL}} \end{aligned}$$

- (b) Use $Q_1C_1 = Q_2C_2$ to solve

First, calculate concentration of iron(II) sulfate in 43.33 mL

$$\frac{1 \text{ g}}{43.33 \text{ mL}} \times 100\% = \mathbf{2.3\% \text{ w/v}}$$

Then apply the above equation

$$Q_2 = \frac{43.33 \times 2.3\%}{1\%} = \mathbf{99.66 \text{ mL}}$$

Then solve for the volume of 0.9% w/v sodium chloride that is needed:

$$99.66 \text{ mL} - 43.33 \text{ mL} = \mathbf{56.33 \text{ mL}}$$

Therefore, 99.66 mL of 1% w/v solution can be prepared and the answer is that 56.33 mL of 0.9% w/v sodium chloride is required to produce it.

(c) There is 1 gram of FeSO_4 and therefore 1 gram of Fe^{2+}

$$\text{mEq} = \frac{\text{mass in mg} \times \text{valence}}{\text{molecular mass}} = \frac{1000 \text{ mg} \times 2}{151.91 \text{ g/mol}} = \mathbf{13.17 \text{ mEq}}$$

(d) There is 1 gram of Fe^{2+} in 43.33 mL of the isotonic solution:

$$x \text{ g} = \frac{1 \text{ g}}{43.33 \text{ mL}} \times 10 \text{ mL} = \mathbf{0.23 \text{ g}}$$

(e) Mass of Base is 151.91 g/mol, find mass of salt

$$\text{Mass of salt form} = 151.91 + 7 \times (2 \times 1.00) + 7 \times 16 = 277.91 \text{ g/mol}$$

$$\text{Active drug moiety} = \frac{151.91 \frac{\text{g}}{\text{mol}}}{277.91 \frac{\text{g}}{\text{mol}}} = \mathbf{0.55}$$

(f)

$$\frac{300 \text{ mg} \times 277.91 \frac{\text{g}}{\text{mol}}}{151.91 \frac{\text{g}}{\text{mol}}} = \mathbf{548.83 \text{ mg}}$$

Question 4 Solution:

(a) Calculate the total amount of drug needed and how many tablets that is required

$$80 \text{ tablets} \times 60 \text{ mg} = \mathbf{4800 \text{ mg}}$$

$$4800 \text{ mg} \div 80 \text{ mg (tablets)} = \mathbf{60 \text{ tablets}}$$

(b) **Since 60 tablets are needed exactly, then the full 5650 mg would provide the phenobarbital required.**

(c)

$$120 \text{ mg} \times 80 \text{ tablets} = 9600 \text{ mg total weight}$$

$$9600 \text{ mg} - 5650 \text{ mg} = \mathbf{3950 \text{ mg}}$$

(d)

$$0.44 \times 60 \text{ mg} = 1 \times \text{Dose}_{\text{injection}}$$

$$\text{Dose}_{\text{injection}} = \mathbf{26.4 \text{ mg}}$$

$$Dose_{injection} \text{ in mL} = 26.4 \text{ mg} \div \frac{30 \text{ mg}}{\text{mL}} = \mathbf{0.88 \text{ mL}}$$

(e)

$$\text{cost of 60mg tablets} = \frac{\$15.46}{100 \text{ tablets}} \times 80 \text{ tablets} = \mathbf{\$12.37}$$

$$\text{cost of 80mg tablets} = \frac{\$18.76}{100 \text{ tablets}} \times 60 \text{ tablets} = \mathbf{\$11.26}$$

The 80-mg tablets are cheaper than the 60 mg tablets.

Question 5 Solution:

(a) Calculate the sensitivity.

$$\text{Sensitivity} = \frac{30}{30 + 12} = \mathbf{0.71}$$

(b) Calculate the specificity

$$\text{Specificity} = \frac{43}{43 + 5} = \mathbf{0.90}$$

(c) Calculate the minimum weighable quantity of the scale.

$$MWQ = \frac{\text{Sensitivity}}{5\%} = \frac{0.3 \text{ mg}}{0.05} = \mathbf{6 \text{ mg}}$$

(d) First find the difference in errors

$$\text{Mass of error} = 41 - 35 = \mathbf{6 \text{ mg}}$$

Next, calculate the percent of error.

$$\frac{6 \text{ mg} \times 100\%}{35 \text{ mg}} = 17.1 \%$$

Question 6 Solution

(a) Multiply the daily dosage by the patient's mass

$$\frac{25 \text{ mg}}{\text{kg}} \times 95 \text{ kg} = 2375 \text{ mg}$$

(b) Calculate both the minimum and maximum monthly dosages

$$\text{Minimum dosage} = 3 \text{ mg} \times 12 = 36 \text{ mg}$$

$$\text{Maximum dosage} = 3 \text{ mg} \times 18 = 54 \text{ mg}$$

(c) First calculate the quantity when escitalopram is administered alone

$$20 \text{ mg} \times 0.67 = 13.4 \text{ mg}$$

Now calculate it when it is co-administered with valproic acid

$$13.4 \text{ mg} \times (1 - 0.15) = 11.39 \text{ mg}$$

(d) Convert hours into minutes and solve for the elimination constant

$$K_{el} = \frac{\ln 2}{t_{\frac{1}{2}}} = \frac{\ln 2}{180 \text{ mins}} = \frac{3.85 \times 10^{-3}}{\text{mins}}$$

(e) Substitute the values to find the amount of drug left after 75 minutes.

$$C_t = 6 \text{ mg} \times e^{-0.00385 \text{ mins}^{-1} \times 75 \text{ minutes}} = 4.49 \text{ mg}$$

(f) Substitute the values into the volume of distribution formula

$$Vd = \frac{D}{C_p} = \frac{2375 \text{ mg}}{123 \frac{\text{mg}}{\text{mL}}} = 19.31 \text{ mL}$$

Question 7 Solution

(a) Calculate the dissociation factor on the basis of 80% dissociation and assuming 100 particles

160 $\text{C}_{18}\text{H}_{37}\text{N}_5\text{O}_9$ ions

400 H_2SO_4

Total 560 ions

$$i = \frac{560}{100} = 5.6$$

Now calculate the E-value:

$$E \text{ value} = \frac{58.5}{1.8} \times \frac{5.6}{1425.45} = 0.13$$

(b) Calculate the lower range of potency

$$250 \text{ mg} \times \frac{564 \text{ mcg}}{1 \text{ mg}} \times \frac{1 \text{ mg}}{1000 \text{ mcg}} = 141 \text{ mg}$$

Calculate the upper range of potency

$$250 \text{ mg} \times \frac{723 \text{ mcg}}{1 \text{ mg}} \times \frac{1 \text{ mg}}{1000 \text{ mcg}} = \mathbf{180.75 \text{ mg}}$$

(c) Find the E-value of diclofenac sodium

$$E \text{ value} = \frac{58.5}{1.8} \times \frac{1.8}{318.13} = \mathbf{0.18}$$

(d) Find the E-value of ketotifen fumarate

$$E \text{ value} = \frac{58.5}{1.8} \times \frac{1.8}{424.498} = \mathbf{0.14}$$

(e) Calculate the sodium chloride equivalent of each compound

$$\text{Tobramycin sulfate equivalent} = 250 \text{ mg} \times 0.13 = \mathbf{32.5 \text{ mg}}$$

$$\text{Diclofenac sodium equivalent} = 75 \text{ mg} \times 0.18 = \mathbf{13.5 \text{ mg}}$$

$$\text{ketotifen fumarate equivalent} = 25 \text{ mg} \times 0.14 = \mathbf{3.5 \text{ mg}}$$

The total amount of sodium chloride in the formula is:

$$32.5 \text{ mg} + 13.5 \text{ mg} + 3.5 \text{ mg} + 800 \text{ mg} = \mathbf{849.5 \text{ mg}}$$

In 100 mL of water, 900 mg of sodium chloride would make the solution isotonic and so there is not enough here.

(f) Conversion factor: $30/100 = 0.3$

$$\text{Converted tobramycin sulfate} = 250 \text{ mg} \times 0.3 = \mathbf{75 \text{ mg}}$$

$$\text{Converted diclofenac sodium} = 75 \text{ mg} \times 0.3 = \mathbf{22.5 \text{ mg}}$$

$$\text{Converted ketotifen fumarate} = 25 \text{ mg} \times 0.3 = \mathbf{7.5 \text{ mg}}$$

$$\text{Converted sodium chloride} = 800 \text{ mg} \times 0.3 = \mathbf{240 \text{ mg}}$$

Question 8 Solution

(a) First calculate the pH using the Henderson-Hasselbalch equation:

$$pH = pKa - \log \frac{[HA]}{[A^-]} = 4.76 - \log \frac{0.25 \text{ M}}{0.2 \text{ M}} = 4.66$$

Calculate the pOH by subtracting the pH from 14

$$pOH = 14 - 4.66 = 9.34$$

(b) The molar ratio is simply salt/acid

$$\text{Molar Ratio} = \frac{\text{salt}}{\text{acid}} = \frac{0.2}{0.25} = 0.8$$

(c) Adding 0.05 M of strong base will convert 0.05 M of base into acid

$$pH = 4.76 - \log \frac{0.2}{0.25} = 4.86$$

(d) Adding 1 M of acid will convert 1 M of acid into base

$$pH = 4.76 - \log \frac{0.35 M}{0.1 M} = 4.22$$

(e) The sum of pKa and pKb is 14

$$pKb = 14 - pKa = 14 - 4.76 = 9.24$$

Question 9 Solution

(a) A 50 kg adult needs 40 mL of compound

$$7 \frac{mg}{mL} \times 40 mL = 280 mg$$

$$\frac{280 mg}{50 mg capsules} = 5.6 tablets \approx 6 tablets$$

(b) Divide the daily dose by the drug concentration

$$\frac{60 mg}{7 \frac{mg}{mL}} = 8.57 mL$$

(c) Calculate pediatric dose using Young's rule

$$Pediatric dose = 7 \frac{mg}{mL} \times \frac{8}{20} = 2.8 \frac{mg}{mL}$$

Calculate pediatric dose using Clark's rule

$$\text{Pediatric dose} = 7 \frac{\text{mg}}{\text{mL}} \times \frac{30 \text{ kg} \times 2.2}{150} = 3.08 \frac{\text{mg}}{\text{mL}}$$

(d) Using the smaller of the two doses in the previous question to calculate the number of capsules needed

$$2.8 \frac{\text{mg}}{\text{mL}} \times 30 \text{ mL} = 84 \text{ mg}$$

$$\frac{84 \text{ mg}}{50 \text{ mg capsules}} = 1.68 \approx 2 \text{ tablets}$$

Question 10 Solution

(a) Calculate the amount of acid needed

$$55 \text{ mL} \times \frac{10 \text{ g}}{100 \text{ mL}} = 5.5 \text{ g acid needed}$$

$$5.5 \text{ g acid} \times \frac{100 \text{ g concentrated acid}}{60 \text{ g acid}} = 9.17 \text{ g concentrated acid}$$

$$9.17 \text{ g} \times \frac{1 \text{ mL}}{1.56} = 5.88 \text{ mL concentrated acid}$$

(b) find the mL of concentrated acid

$$5.5 \text{ g} \times \frac{100 \text{ mL concentrated acid}}{60 \text{ g acid}} = 9.17 \text{ mL concentrated acid}$$

(c) Find the error and then calculate the percent error

$$\text{Error} = 9.17 \text{ mL} - 5.89 \text{ mL} = \mathbf{3.28 \text{ mL}}$$

$$\text{Percent error} = \frac{3.28 \text{ mL} \times 100}{5.89 \text{ mL}} = \mathbf{55.69 \%}$$

(d) Calculate the new percent error

$$9.17 \text{ mL concentrated acid} \times 1.56 \frac{\text{g}}{\text{mL}} = \mathbf{14.31 \text{ g concentrated acid}}$$

$$14.31 \text{ g concentrated acid} \times \frac{60 \text{ g acid}}{100 \text{ g concentrated acid}} = \mathbf{8.59 \text{ g acid}}$$

$$\frac{8.59 \text{ g acid}}{55 \text{ mL solution}} \times 100\% = \mathbf{15.62\% w/v}$$

Question 11 Solution

(a) First calculate the units in the heparin bolus dose

$$\frac{60 \text{ units}}{\text{kg}} \times 89 \text{ kg} = \mathbf{5340 \text{ units}}$$

Now calculate milliliters from units

$$5340 \text{ units} \div \frac{1000 \text{ units}}{\text{mL}} = \mathbf{5.34 \text{ mL}}$$

(b) Find the revised rate by multiplying the initial rate by 1.4 (increase in 40% = 1 + .4)

$$\frac{15 \text{ units}}{\frac{\text{kg}}{\text{h}}} \times 1.4 = \frac{\mathbf{21 \text{ units}}}{\frac{\text{kg}}{\text{h}}}$$

(c) Calculate the units per h

$$\frac{21 \text{ units}}{\frac{kg}{h}} \times 89 \text{ kg} = \frac{1869 \text{ units}}{h}$$

Next calculate the number of milliliters delivered per hour

$$\frac{1869 \text{ units}}{h} \times \frac{100 \text{ mL}}{5000 \text{ units}} = 37.38 \frac{\text{mL}}{h}$$

Finally, calculate the drops per hour

$$\frac{37.38 \text{ mL}}{h} \times \frac{20 \text{ drops}}{\text{mL}} = \frac{747.6 \text{ drops}}{h}$$

Question 12 Solution

(a) Calculate the total volume needed

$$\text{total volume} = 24 \text{ bottles} \times 25 \text{ mL} = 600 \text{ mL}$$

Calculate the formula conversion factor

$$\text{formula conversion factor} = \frac{600 \text{ mL}}{100 \text{ mL}} = 6$$

Multiply every value in the formula by the conversion factor

$$\text{Testosterone} = 1.5 \text{ g} \times 6 = 9 \text{ g}$$

$$\text{Alcohol} = 20 \text{ mL} \times 6 = 120 \text{ mL}$$

$$\text{Propylene glycol} = 25 \text{ mL} \times 6 = 150 \text{ mL}$$

$$\text{Benzalkonium chloride} = 10 \text{ mg} \times 6 = \mathbf{60 \text{ mg}}$$

$$\text{Purified water} = 100 \text{ mL} \times 6 = \mathbf{600 \text{ mL}}$$

(b) Determine the amount of stock solution that would provide 60 mg

$$60 \text{ mg} \times \frac{1 \text{ g}}{1000 \text{ mg}} \times \frac{250 \text{ mL}}{1 \text{ g}} = \mathbf{15 \text{ mL}}$$

(c) Find how much propylene glycol found in each bottle

$$\frac{150 \text{ mL}}{24 \text{ bottles}} = \mathbf{6.25 \frac{\text{mL}}{\text{bottle}}}$$

Find the amount of foreign substance found in each bottle

$$6.25 \text{ mL (propylene glycol)} \times \frac{2.3 \text{ g (foreign substance)}}{1,000,000 \text{ mL}} = 1.44 \text{ g} \times 10^{-5} = \mathbf{0.0144 \text{ mg}}$$

(d) Calculate the error weight then the percent error

$$\text{Error} = 10.36 \text{ g} - 9 \text{ g} = \mathbf{1.36 \text{ g}}$$

$$\text{Percent Error} = \frac{1.36 \text{ g}}{9 \text{ g}} \times 100 \% = \mathbf{15.1 \%}$$

(e) Calculate active drug moiety of testosterone cypionate

$$\frac{288.4}{412.6} = \mathbf{0.7}$$

Calculate testosterone cypionate equivalent required

$$\frac{9 \text{ g}}{0.7} = \mathbf{12.86 \text{ g}}$$

Find mL required

$$12.86 \text{ g} \times \frac{1 \text{ mL}}{300 \text{ mg}} \times \frac{1000 \text{ mg}}{1 \text{ g}} = \mathbf{42.87 \text{ mL}}$$

(f) Calculate the amount of testosterone

$$\frac{53.29 \text{ ng}}{10 \text{ mL}} \times \frac{1000 \text{ mL}}{\text{L}} \times \frac{1 \text{ L}}{10 \text{ L}} = \mathbf{\frac{532.9}{\text{dL}}}$$

The testosterone levels are within normal range.

Question 13 Solution

(a) 0.2% of every spray is oxymetazoline hydrochloride

$$0.162 \text{ mL} \times 0.002 = 0.00324 \text{ g} = \mathbf{0.324 \text{ mg}}$$

(b) Calculate the percent strength

$$\frac{75 \text{ mg}}{\text{mL}} \times \frac{\text{g}}{1000 \text{ mg}} = \frac{0.075 \text{ g}}{\text{mL}}$$

$$\frac{0.075 \text{ g}}{1 \text{ mL}} = \frac{x}{100 \text{ mL}}$$

$$x = \mathbf{7.5\%}$$

Calculate the ratio strength

$$\frac{100 \text{ mL}}{7.5 \text{ g}} = 13.33$$

(c) Calculate the volume of 225 sprays

$$0.162 \text{ mL} \times 225 \text{ sprays} = 36.45 \text{ mL}$$

Subtract total volume of bottle by volume of sprays

$$45 \text{ mL} - 36.45 \text{ mL} = 8.55 \text{ mL}$$

(d) 4 sprays required every day and a container contains 250 sprays

$$\frac{250 \text{ sprays}}{4 \text{ sprays per day}} = 62.5 \text{ days}$$

Round down and the bottle will last 62 days.

Question 14 Solution

(a) Calculate the number of capsules that are required

$$0.2 \text{ mCi} \times \frac{37 \text{ MBq}}{\text{mCi}} \times \frac{1 \text{ capsule}}{3.7 \text{ MBq}} = 2 \text{ capsules}$$

(b) Find the radioactivity decay constant using half-life

$$\lambda = \frac{\ln 2}{t_{\frac{1}{2}}} = \frac{\ln 2}{13.2 \text{ h}} = 0.0525 \text{ h}^{-1}$$

Solve for amount of activity after 30 hours

$$N = N_0 \times e^{-\lambda t} = 0.2 \times e^{-0.0525 \times 30} = \mathbf{0.04 \text{ mCi}}$$

(c) Rearrange the formula for N_0

$$N_0 = \frac{N}{e^{-\lambda t}}$$

Solve for N_0

$$N_0 = \frac{0.09}{e^{-0.0525 \times 18}} = \mathbf{0.232 \text{ mCi}}$$

Question 15 Solution

(a) The molecular formula for tramadol hydrochloride is $\text{HCl}(\text{C}_{16}\text{H}_{25}\text{NO}_2)$

$$\text{Total molar mass} = 36.46 \frac{\text{g}}{\text{mol}} (\text{HCL}) + 263.381 \frac{\text{g}}{\text{mol}} (\text{tramadol}) = \mathbf{299.84 \frac{\text{g}}{\text{mol}}}$$

Calculate volume required from 10% w/v

$$30 \text{ mEq} \times \frac{299.84 \text{ mg}}{\text{mEq}} \times \frac{1 \text{ g}}{1000 \text{ mg}} \times \frac{100 \text{ mL}}{10 \text{ g}} = \mathbf{89.95 \text{ mL}}$$

(b) Calculate the total volume

$$\text{Total Volume} = 1000 \text{ mL (dextrose)} + 89.95 \text{ mL (tramadol hydrochloride)} = \mathbf{1089.95 \text{ mL}}$$

Find the mOsmol/L of tramadol hydrochloride

$$\frac{30 \text{ mEq}}{1089.95 \text{ mL}} \times \frac{299.98 \text{ mg}}{\text{mEq}} \times \frac{2 \text{ mOsmol}}{299.98 \text{ mg}} \times \frac{1000 \text{ mL}}{\text{L}} = \mathbf{\frac{55.05 \text{ mOsmol}}{\text{L}}}$$

Find the mOsmol/L of dextrose (m.w = 180)

$$\frac{50 \text{ g}}{1089.95 \text{ mL}} \times \frac{1000 \text{ mg}}{\text{g}} \times \frac{1 \text{ mOsmol}}{180 \text{ mg}} \times \frac{1000 \text{ mL}}{\text{L}} = \mathbf{\frac{254.86 \text{ mOsmol}}{\text{L}}}$$

Calculate the total osmolarity of the IV solution

$$254.86 + 55.05 = \mathbf{\frac{309.91 \text{ mOsmol}}{\text{L}}}$$

(c) Calculate mEq based on 24 hours (1 day)

$$\frac{30 \text{ mEq}}{1089.94 \text{ mL}} \times \frac{30 \text{ mL}}{\text{h}} \times \frac{24 \text{ h}}{\text{day}} = \mathbf{\frac{19.82 \text{ mEq}}{\text{day}}}$$

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CHANGE LOG**April 2026**

- Updated example 66, pages 42-43

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